Instruction Routing in Fine-Tuned Vision-Language-Action Models: A Cross-Architecture Analysis

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Abstract: When vision-language models are fine-tuned for robotic action prediction, which representational changes reflect training and which are inherited from the pretrained backbone? We conduct a cross-architecture representational analysis of three VLAs (OpenVLA, TracVLA, and NORA) across four LIBERO task suites, comparing trained and reference checkpoints. Using layer-wise action probes, attention profiling, and counterfactual instruction perturbations, we find that the two tested VLM-to-VLA transitions exhibit pronounced peak-to-exit reductions in action-probe decodability, whereas the tested VLA-to-VLA transition shows only small fluctuations. An NLP finetuning control on each of the three backbone families ($\Delta R^2 = 0.000$) is consistent with exit collapse being specific to action-oriented training rather than generic fine-tuning. Trained action-producing VLAs show medium-to-large sensitivity to instruction perturbations ($d = 0.45$–1.52), while L1 instruction-dominant attention largely pre-exists in the pretrained backbones. These results provide a cross-architecture characterization of instruction-related representations in action-finetuned VLAs and clarify which observed patterns are training-associated versus inherited from reference models.

Keywords: Vision-Language-Action models, representational analysis, attention routing, fine-tuning

1 Introduction

Vision-language-action models (VLAs) fine-tune pretrained language models on robot demonstrations to produce actions from visual observations and language instructions [1, 2], but which of the resulting representational changes reflect action training and which are inherited from the pretrained backbone? In these decoder-only architectures, image tokens precede instruction tokens, and the causal attention mask restricts multimodal integration to a single readout token [3, 4]; how representations at this critical position are reshaped by action fine-tuning remains unknown.

Behavioral studies show that VLAs exhibit visual dominance [5] and underweight instructions [6], but existing representational analyses target single architectures. No prior work tests whether instruction routing mechanisms are consistent across backbone families or decomposes training-associated effects from inherited base-model priors.

We address this gap with a cross-architecture analysis of three VLAs (OpenVLA [2], TracVLA [7], and NORA [8]) across four LIBERO [9] task suites (§4.1). TracVLA and NORA baselines are pure VLMs ($N = 2$ independent VLM-to-VLA transitions); OpenVLA's baseline is the OXE-pretrained [10] VLA (VLA-to-VLA transition).

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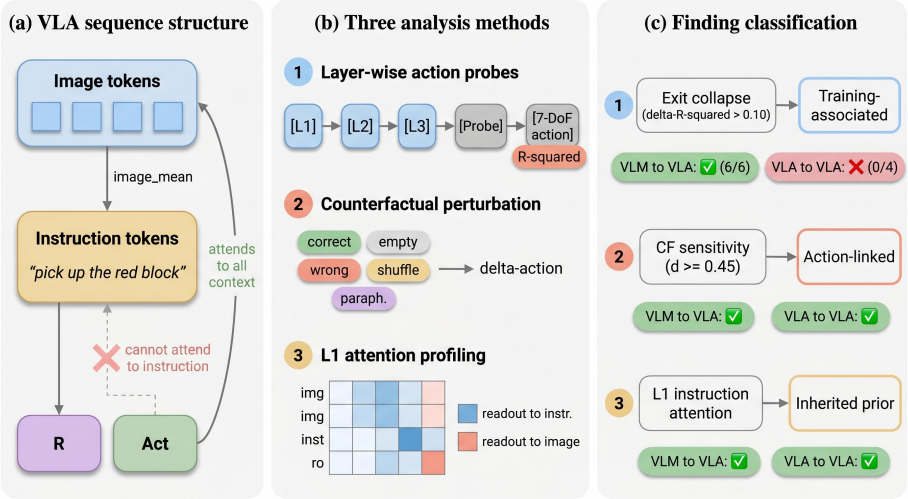


Figure 1: **Analysis framework.** We analyze hidden states at image-token averages and the readout token via layer-wise action probes, counterfactual instruction perturbations, and L1 attention profiling to distinguish training-associated changes from inherited backbone priors.

Our analysis reveals three findings (Figure 1): (1) VLM-to-VLA fine-tuning is associated with substantial R^2 exit collapse ($\Delta R^2 = 0.15$–0.45), whereas VLA-to-VLA fine-tuning and reference models show only small fluctuations ($\Delta R^2 < 0.10$), with three NLP controls (all $\Delta R^2 = 0.000$) confirming transition-type specificity; (2) all trained VLAs exhibit medium-to-large counterfactual sensitivity ($d = 0.45$–1.52); and (3) L1 instruction attention reflects base-model priors, not VLA training (§4.3–4.5). While concurrent work evaluates VLA instruction-following behaviorally [11, 12], we provide complementary representational analysis of how instruction information is routed internally. Our contributions are:

1. A cross-architecture representational analysis of three VLAs, combining trained/reference controls with token-level probing across a 3×4 architecture-suite matrix.
2. Evidence that R^2 exit collapse is associated with VLM-to-VLA transitions ($\Delta R^2 > 0.10$ across two architectures) but not with VLA-to-VLA fine-tuning or NLP instruction tuning of the same backbones.
3. A decomposition of VLA instruction routing into training-associated effects and inherited base-model priors, based on cross-architecture trained/reference contrasts.

2 Related Work

Vision-language-action models. VLAs fine-tune large language models on robot demonstrations to produce actions from visual and language input [1, 2, 13, 14, 7, 8, 15]. NLP instruction tuning is well-characterized behaviorally [16, 17], but the representational effects of VLA fine-tuning remain unexamined. Fine-tuning operates in a low intrinsic dimension [18] and can distort pretrained features [19]; recent VLA-specific work proposes representation-preserving strategies [20, 21]. Existing analyses target single architectures and cannot separate training dynamics from backbone artifacts.

Mechanistic and behavioral analysis. Mechanistic interpretability has revealed residual stream communication [22], induction heads [23], and task circuits [24, 25]; causal tracing localizes factual recall [26]; attention probing shows heads encode syntactic relations [27] and most can be pruned [28]. In VLMs, recent work characterizes visual token redundancy [29], progressive visual-textual alignment [30, 31], and instruction tokens as shallow-layer modality arbitration anchors [32]. Behavioral evaluations of VLAs reveal visual dominance [5, 33, 34]; mechanistic studies identify sparse semantic directions [6], interpretable features [12, 35], symbolic representations [36, 37],

comprehension-action trade-offs [38], and task-specific attention heads for selective fine-tuning [39]. Concurrent continual-learning studies find that pretrained VLAs resist forgetting under sequential fine-tuning [40], consistent with our observation that VLA-to-VLA transitions preserve exit-layer structure. All prior representational analyses target a single architecture; our 3×4 cross-architecture matrix tests whether mechanisms replicate across backbone families.

Concurrent work. Three concurrent studies address complementary aspects of VLA understanding. Fang et al. [11] construct a behavioral benchmark of counterfactual instruction failures on LIBERO; their evaluation measures *whether* the model follows the correct instruction, while our perturbation measures *how much* hidden-state geometry changes (§4.4). Buurmeijer et al. [35] probe low-level motor features in OpenVLA; instruction-level routing is the focus of our analysis. Concurrent work observes visual representation collapse during VLA fine-tuning and proposes alignment strategies [41]; our analysis quantifies a complementary phenomenon (layer-wise action-probe decodability) through ΔR^2 metrics across architectures. We additionally distinguish training-associated from inherited effects across three architectures, a decomposition absent from prior single-architecture studies.

Probing and similarity methodology. RSA [42], CKA [43], and linear probes [44, 45] provide complementary geometric and functional evaluation. We adopt RSA and MLP probes (CKA in appendix); the two metrics can diverge on the same data (Appendix A.5), and Cohen's d vs. cosine distance are not comparable across trained/reference conditions (§4.4).

3 Method

We investigate how action-relevant representations are organized across three VLA architectures using complementary probing, attention, and perturbation analyses.

3.1 Architectures and Data

We study three VLAs that span distinct backbone families (Table 3). **OpenVLA** [2, 13] chains a frozen DinoSigLIP vision backbone [46, 47] (DINOv2 ViT-L/14 + SigLIP-SO400M), an MLP projector, and Llama-2-7B [48] (6.74B parameters, 32 layers, 32 multi-head attention [MHA] heads) for autoregressive 7-D action decoding. It is pretrained on Open X-Embodiment [10] (OXE; 970K episodes across multiple robot platforms). **TracVLA** [7] pairs a SigLIP encoder with Phi-3-mini [49] (3.8B parameters, 32 layers, 32 MHA heads). Spatial trace conditioning renders visual trajectory overlays on input images to provide spatial guidance for action prediction. TracVLA is trained on an OXE subset. **NORA** [8] uses Qwen2.5-VL-3B ($\sim$3B parameters, 36 layers, grouped-query attention [GQA] with 16 query heads and 2 KV heads) with the native Qwen2.5-VL vision encoder, which processes images at dynamic resolution without a separate projection stage.

Each architecture is evaluated in trained and reference conditions; we define the *reference checkpoint* as the corresponding pretrained model before LIBERO fine-tuning. A critical asymmetry distinguishes the reference baselines: the OpenVLA reference is an OXE-pretrained VLA, already fine-tuned for action prediction on 970K episodes, while the TracVLA and NORA references are pure vision-language models (Phi-3-Vision and Qwen2.5-VL) with no action training. OpenVLA trained-vs-reference comparisons therefore measure a VLA-to-VLA transition; TracVLA and NORA comparisons measure VLM-to-VLA transitions. This asymmetry affects the interpretation of all trained/reference comparisons throughout the paper. For OpenVLA, two additional conditions support deep-dive analyses (§4.2): the Llama-2 base language model (no VLA fine-tuning) and a random-weight initialization via `from_config()`.

We evaluate on LIBERO [9], comprising four task suites (Goal, Spatial, Object, Long-Horizon) with 10 tasks and 50 demonstrations each. The three architectures and four suites form a 3×4 matrix; 10 of 12 cells are filled (NORA Object and Long-Horizon lack available checkpoints).

Architecture details are summarized in Table 3 (Appendix A.1).

3.2 Representation Extraction

From each architecture, we extract hidden states at every transformer layer from two token positions: (a) **image_mean**, the mean activation over image tokens, and (b) **readout**, the hidden state at the final pre-action token, which follows both image and instruction tokens in the causal sequence and can attend to all preceding context. L0 denotes the embedding layer output, L1 through LN the hidden states after each transformer layer ($N = 32$ for OpenVLA and TracVLA, $N = 36$ for NORA), with the final layer including the output layer norm. In all three architectures, image tokens precede instruction tokens; the causal mask prevents image tokens from attending to instructions by construction.

We cross two factors in a 2×2 factorial design: *training status* (trained vs. reference) and *instruction presence* (with vs. without task description), applied to all 10 available cells in the cross-architecture matrix. OpenVLA additionally provides representations from the DinoSigLIP projector output (4,096-D), DINOv2 and CLIP external baselines, and the branch-split before fusion; these sources support the OpenVLA deep-dive analyses (§4.2).

3.3 Cross-Architecture Analyses

Three analyses are applied to every available cell in the 3×4 matrix.

Action probes. To quantify how much action-relevant structure is accessible at each layer, we train two-layer MLP probes [44] to predict 7-DoF action vectors from frozen representations (architecture and training details in Appendix A.2). We report probe R^2 as a measure of action decodability. Higher R^2 indicates that actions are more accurately predictable from that representation under the probe; it should not be interpreted as evidence that the model causally uses that layer for action generation. We define ΔR^2 as the difference between the maximum layer-wise R^2 and the exit-layer R^2, and use $\Delta R^2 > 0.10$ as a descriptive threshold for substantial peak-to-exit reduction. Train/test splits follow demonstration boundaries to prevent within-trajectory leakage. Demo-level splitting reduces R^2 from 0.7–0.9 to 0.4–0.6 compared to random splitting, confirming that temporal correlations inflate aggregate estimates. The qualitative separation between VLM-to-VLA and other conditions is robust to threshold choice: any threshold in $(0.097, 0.156)$ yields identical grouping (Appendix A.19.7). 95% CIs are obtained via Monte Carlo cross-validation with demo-level splitting ($n = 1{,}000$ iterations): the canonical peak layer is fixed from the full-data split to avoid winner's-curse bias, and entire demonstrations are resampled without replacement to respect nested temporal dependencies. Linear probes yield consistent patterns (Appendix A.14), confirming that observed effects are not artifacts of MLP probe capacity.

L1 attention head profiling. To determine whether attention heads preferentially route instruction information to the readout position, each L1 head is classified as instruction-dominant ($> 50\%$ of readout-token attention to instruction tokens) or image-dominant, reported as k/N heads (32 for MHA, 16 for GQA). Sensitivity analysis at thresholds from 30% to 70% confirms ordering robustness (Appendix A.22.8).

Counterfactual instruction perturbation. To test whether the model's actions depend on instruction content, the instruction is replaced with four conditions: empty string, wrong task, shuffled tokens, and paraphrased. We measure trained-model sensitivity via Cohen's d on action MSE (OpenVLA, TracVLA) or cosine distance (NORA); reference VLMs use cosine distance on hidden representations. These metrics operate on different scales; cross-metric ratios are invalid (§4.4).

OpenVLA additionally supports cross-instruction probes, representational similarity analysis (RSA) [42], and external baselines (DINOv2 [47], CLIP [50]); these single-architecture methods are detailed in Appendix A.2.

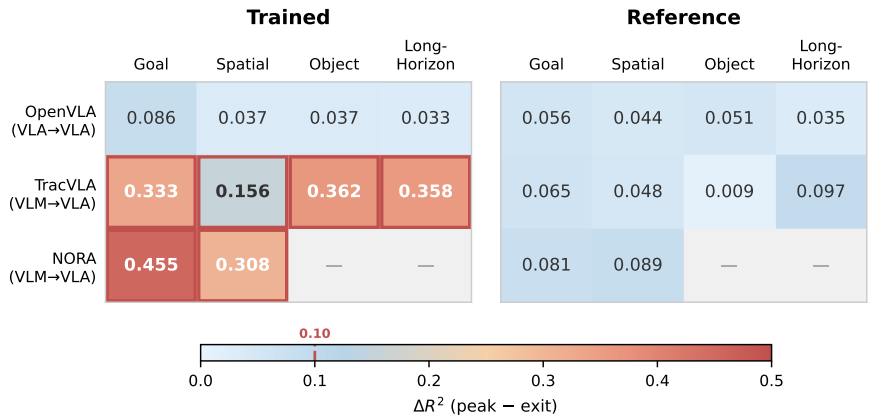


Figure 2: **Cross-architecture evaluation matrix.** Each cell reports ΔR^2 (peak−exit). OpenVLA: VLA-to-VLA; TracVLA and NORA: VLM-to-VLA. Gray cells: checkpoints not released by original authors. Red borders indicate $\Delta R^2 > 0.10$.

4 Experiments

4.1 Experimental Setup

We apply all three analyses (§3.3) to the 3×4 architecture-suite matrix described in §3.1 (Table 1). Of the 12 cells, 10 are filled; NORA Object and Long-Horizon lack available checkpoints. For OpenVLA LIBERO-Goal, $N = 63{,}228$ observations; other cells are at comparable scale. For trained models that produce action outputs, counterfactual sensitivity is measured via Cohen's d on action MSE. For reference VLMs (Phi-3-Vision, Qwen2.5-VL) that do not produce actions, sensitivity is measured via cosine distance on hidden representations; these metrics operate on different scales and are not directly comparable (§4.4). OpenVLA's OXE-pretrained checkpoint produces actions, permitting same-metric (Cohen's d) comparison in that condition. Bonferroni correction details appear in Appendix A.30. Figure 2 summarizes the 3×4 architecture-suite matrix.

4.2 OpenVLA: Motivating Single-Architecture Analysis

Before testing cross-architectural generality, we establish three core phenomena on OpenVLA [2] (full analysis in Appendix A.3). At L1, three conditions form a gradient: random-weight initialization allocates 0/32 instruction-dominant heads, Llama-2 base [48] allocates 9/32, and VLA-trained allocates 24/32, indicating that VLA fine-tuning shapes instruction routing beyond language-model priors. Counterfactual perturbation produces graded action divergence ordered by semantic distance ($d = 0.81$–1.18 across Goal and Spatial suites). Dose-response ablation of instruction-dominant heads yields monotonic trajectory divergence ($d = 0.48$–0.64; Appendix A.12). Ablation redirects action outputs ($d = 0.65$) without increasing ground-truth error ($\Delta\text{MSE} = -0.001$, n.s.; Appendix A.12), consistent with instruction heads controlling direction selection rather than action precision. R^2 probing localizes training effects at the readout token ($+0.069$) rather than image_mean ($+0.015$), and readout instruction classification remains at 100% through L31 in the trained model (reference: 82% decay). These phenomena motivate the cross-architecture tests that follow.

4.3 Exit Collapse: Layer-Wise Action Probing

We define R^2 exit collapse as $\Delta R^2 = R^2_{\text{peak}} - R^2_{\text{exit}}$, the drop in action-probe decodability from peak layer to exit layer (Table 1). VLM-to-VLA fine-tuning is associated with substantial exit collapse across all tested conditions: TracVLA [7] $\Delta R^2 = 0.156$–0.362 across all four suites, and NORA [8] $\Delta R^2 = 0.308$ (Spatial) and 0.455 (Goal) (Figure 3), all exceeding the $\Delta R^2 > 0.10$ threshold for substantive collapse (§3.3). Under VLA-to-VLA conditions, OpenVLA [2] shows only small fluctuations across all four suites ($\Delta R^2 = 0.033$–0.086, all below 0.10); reference baselines similarly show $\Delta R^2 \leq 0.097$. The gap between the largest non-collapse value (0.097) and smallest

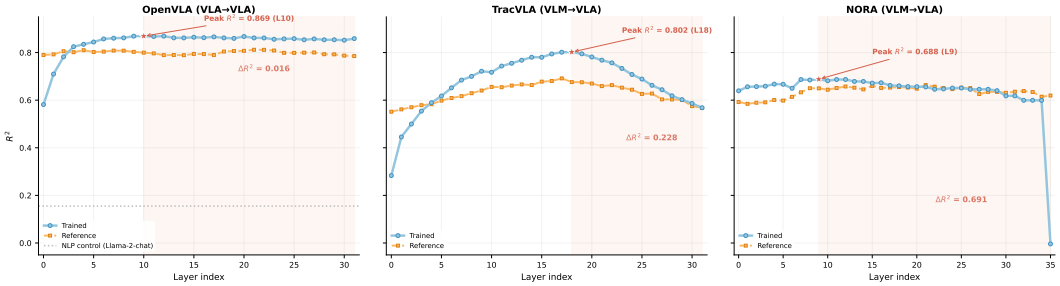


Figure 3: **Layer-wise action-probe R^2.** Solid: trained; dashed: reference. VLM-to-VLA transitions show pronounced peak-to-exit drops ($\Delta R^2 = 0.16$–0.46); VLA-to-VLA remains flat (< 0.09).

Table 1: **Peak-to-exit action-probe drop across the architecture-suite matrix.** ΔR^2 is computed as the difference between the maximum layer-wise action-probe R^2 and the exit-layer R^2. Values above 0.10 are treated as substantial descriptive drops. OpenVLA is a VLA-to-VLA comparison, whereas TracVLA and NORA are VLM-to-VLA comparisons.

			ΔR^2 (peak − exit)			
Architecture	Suite	Transition	Trained	95% CI	Reference	95% CI
OpenVLA	Goal	VLA→VLA	0.086	[−0.060, 0.110]	0.056	[0.010, 0.080]
OpenVLA	Spatial	VLA→VLA	0.037	[0.010, 0.100]	0.044	[0.030, 0.110]
OpenVLA	Object	VLA→VLA	0.037	[0.000, 0.090]	0.051	[0.000, 0.090]
OpenVLA	Long	VLA→VLA	0.033	[0.000, 0.100]	0.035	[0.030, 0.100]
TracVLA	Goal	VLM→VLA	0.333	[0.210, 0.450]	0.065	[−0.110, 0.050]
TracVLA	Spatial	VLM→VLA	0.156	[0.090, 0.220]	0.048	—[a]
TracVLA	Object	VLM→VLA	0.362	[0.280, 0.500]	0.009	[−0.070, 0.020]
TracVLA	Long	VLM→VLA	0.358	[0.170, 0.530]	0.097	[−0.020, 0.110]
NORA	Goal	VLM→VLA	0.455[b]	—[b]	0.081	[−0.010, 0.110]
NORA	Spatial	VLM→VLA	0.308	[0.300, 0.510]	0.089	[−0.010, 0.090]

NORA Object and Long-Horizon checkpoints are unavailable.
[a]CI artifact from flat R^2 profile. [b]Bootstrap CI unreliable (exit-layer $R^2 \approx -0.006$).

collapse value (0.156) spans 0.059; any threshold in this interval yields identical grouping, and a permutation test rejects random label assignment ($p = 0.008$; Appendix A.19.7). Monte Carlo cross-validation supports this separation: VLM-to-VLA trained CIs lie entirely above zero (lower bounds ≥ 0.09), while VLA-to-VLA trained CIs span or nearly span zero (Table 1).

NLP finetuning control. To test whether exit collapse is specific to action-oriented training, we probe three chat-finetuned models that share backbones with our VLAs but lack action heads or robot data: Llama-2-7b-chat (OpenVLA), Phi-3-mini-4k-instruct (TracVLA), and Qwen2.5-3B-Instruct (NORA).[1] All three produce flat R^2 profiles ($\Delta R^2 = 0.000$; $R^2 = 0.155 \pm 0.001$, 0.155 ± 0.000, and 0.155 ± 0.000 respectively; Appendix A.34), contrasting with VLM-to-VLA collapse ($\Delta R^2 = 0.156$–0.455). The absence of exit collapse in any NLP control is inconsistent with exit collapse being a generic consequence of language-model fine-tuning. **In sum, exit collapse is consistently associated with VLM-to-VLA transitions and absent in both VLA-to-VLA and NLP-only fine-tuning.**

[1]Qwen2.5-3B-Instruct shares the Qwen2.5 language backbone with NORA's Qwen2.5-VL-3B but lacks the vision encoder and cross-attention modules; see Appendix A.34.

Table 2: **Counterfactual sensitivity summary.** Cohen's d range and most sensitive perturbation for action-producing models (OpenVLA, TracVLA). NORA uses cosine distance on hidden representations (different scale; see §4.4.2).

Architecture	d range	Most sensitive	Metric
OpenVLA	0.45–1.21	empty (2/4 suites)	Cohen's d
TracVLA	0.97–1.52	empty (4/4 suites)	Cohen's d
NORA	0.28–sat.	—	cosine dist.[a]

[a] Representation-space only; not comparable to Cohen's d (§4.4.2).
Empty ranks first in 6/8 suites total across both architectures.

4.4 Instruction Sensitivity: Counterfactual Perturbation

4.4.1 Instruction Sensitivity in Action-Producing VLAs

We first evaluate counterfactual sensitivity in action-producing models using a shared action-space metric. Across OpenVLA and TracVLA, instruction perturbations produce medium-to-large action divergence in all eight architecture-suite cells (Figure 4).

Every perturbation condition yields $d = 0.45$–1.52 (Table 2; full per-condition rankings in Appendix Table 4). The "empty" condition ranks first in 6 of 8 cells ($d = 1.17$–1.52); "paraphrased" ranks last in 7 of 8, indicating that VLAs encode instruction content beyond token-level matching. TracVLA produces systematically higher effect sizes ($d = 0.97$–1.52) than OpenVLA (0.45–1.21). Removing the instruction entirely often produces larger divergence than replacing it with a wrong task, suggesting that instructions serve as active computation signals rather than task selectors.

Two OpenVLA cells permit same-metric trained/reference comparison: on Spatial, training increases sensitivity (trained−reference difference: $+0.05$ to $+0.24$ across conditions); on Long-Horizon, the OXE-pretrained model exceeds the LIBERO-trained model ($d = 1.05$ vs. 0.57), consistent with the general-purpose model maintaining broader instruction responsiveness. **Instruction perturbation produces graded action divergence across all tested conditions, with effect sizes reflecting semantic distance.**

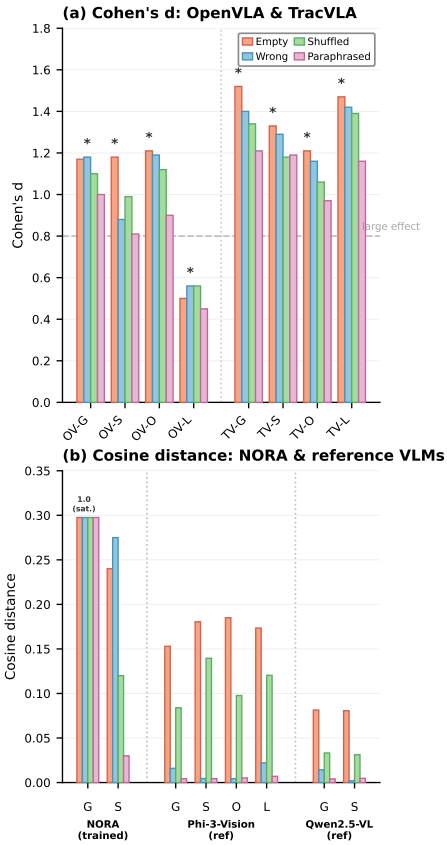


Figure 4: Counterfactual sensitivity across the architecture-suite matrix. Bars show Cohen's d (top perturbation) for action-producing models and cosine distance for representation-space models.

4.4.2 Representation-Space Sensitivity in NORA and Reference VLMs

NORA and reference VLMs lack action outputs; their sensitivity is measured via cosine distance on hidden representations (different scale from Cohen's d). NORA Goal produces saturated cosine distances (1.0); NORA Spatial peaks at 0.275; reference VLMs show weak sensitivity (≤ 0.19; Appendix A.4). The trained-reference contrast is consistent with action-space findings, though the different metric prevents direct magnitude comparison.

4.5 Inherited Routing: L1 Attention Profiling

L1 instruction-dominant head counts across architectures reveal a pattern that contrasts with the training-associated effects documented in §4.3 and §4.4. The TracVLA reference (Phi-3-Vision) allocates 32/32 heads to instruction-dominant attention across all four suites, saturating the capacity of its 32-head L1 before any VLA training. Training slightly reduces this count to 27–30/32, indicating that VLA fine-tuning does not increase L1 instruction-dominant attention in this architecture. The NORA reference (Qwen2.5-VL) allocates 13–14 of its 16 L1 heads to instruction dominance (81–88%), and training leaves this essentially unchanged at 12–13/16. OpenVLA shows 24–26/32 instruction-dominant heads in the OXE-pretrained baseline; trained values are similar (14–24/32), with the largest change in Long-Horizon where training reduces the count from 25 to 14 heads (Figure 5).

If VLA training drove L1 instruction routing, reference VLMs should show low attention that increases after training, as observed in OpenVLA's random-weight condition (0/32; §4.2). Instead, both reference VLMs already saturate or near-saturate, and training leaves these counts unchanged or slightly reduced. These results indicate that L1 instruction-dominant attention is largely inherited from pretrained backbones rather than created de novo by VLA fine-tuning. **L1 instruction routing is an inherited backbone property, not a consequence of VLA training.**

4.6 Synthesis: Training-Associated vs. Inherited Routing

The three analyses separate two empirical patterns: exit collapse and counterfactual sensitivity are training-associated (pronounced in VLM-to-VLA, small in VLA-to-VLA), while L1 instruction attention is inherited from reference backbones. Despite qualitatively different reference profiles, trained OpenVLA and TracVLA converge to nearly identical mid-layer peak decodability ($R^2 = 0.637$ vs. 0.634; Appendix A.29), suggesting convergent representational reorganization. These probes measure decodability, not causal reliance; with this caveat, the consistent three-way pattern provides a first cross-architecture decomposition of VLA instruction routing.

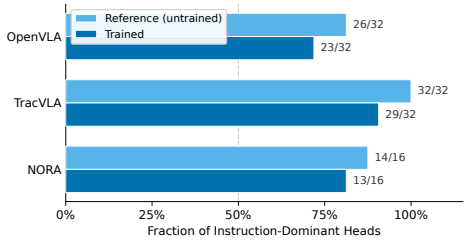


Figure 5: L1 instruction-dominant head counts (trained vs. reference). Medians across available LIBERO suites per architecture.

5 Discussion and Conclusion

Scope and limitations. All findings are correlational (confounded by data distribution and optimization; $N = 2$ VLM-to-VLA transitions; Appendix A.35); we analyze causal-decoder VLAs on LIBERO only (flow-matching [14] and diffusion [51] architectures may differ); all analyses are open-loop. Trained ΔR^2 correlates with published success rates ($\rho = 0.583$, $p = 0.10$; Appendix A.33) but is underpowered ($n = 9$; effect size is large, $|\rho| > 0.5$); tracing collapse onset remains for future work.

Practical implications. ΔR^2 provides a cheap diagnostic (> 0.10 signals reorganization); preserved VLA-to-VLA exit structure justifies staged pipelines; mid-layer probes outperform exit-layer probes where collapse is pronounced (Appendix A.33).

Conclusion. Across three architectures and four LIBERO suites, VLM-to-VLA fine-tuning produces exit collapse ($\Delta R^2 = 0.15$–0.45) while VLA-to-VLA conditions remain flat (< 0.10). All trained VLAs show graded counterfactual sensitivity ($d = 0.45$–1.52); L1 instruction attention is inherited from pretrained backbones; NLP controls ($\Delta R^2 = 0.000$) confirm transition-type specificity, providing reusable diagnostics for future VLA evaluation and training pipeline design. Code and data will be released upon acceptance.

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Use of Generative AI. Generative AI tools (Claude, GPT-4) assisted with code implementation and manuscript editing; all scientific content was produced by the authors.

A Supplementary Material

Unless noted otherwise, all analyses use OpenVLA-7B (`openvla/openvla-7b` on HuggingFace) evaluated on LIBERO-Goal (10 tasks $\times$ 50 demonstrations, $N = 63{,}228$ observations after excluding terminal frames). Model conditions, representation extraction, metric definitions, and probe architectures follow §3. Cross-suite experiments (§A.18) use LIBERO-Spatial with the same protocol (10 tasks $\times$ 50 demonstrations). All experiments ran on 2$\times$ NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each); random seed is 42 unless stated otherwise.

A.1 Architecture Summary

Table 3: Architectures studied. The reference baselines differ in kind: the OpenVLA reference is an OXE-pretrained VLA that produces actions; TracVLA and NORA references are pure VLMs with no action training.

Arch.	Backbone	Layers	Attention	Vision Enc.	Reference Baseline
OpenVLA	Llama-2-7B	32	32 MHA	DinoSigLIP $\rightarrow$ MLP proj.	OXE-pret. VLA
TracVLA	Phi-3-mini	32	32 MHA	SigLIP + spatial trace	Phi-3-Vision (VLM)
NORA	Qwen2.5-VL-3B	36	16q/2kv GQA	Qwen2.5-VL native	Qwen2.5-VL (VLM)

A.2 OpenVLA-Specific Analysis Methods

Three additional analyses are applied to OpenVLA to characterize instruction encoding and representational geometry.

Cross-instruction probes. Each visual frame is paired with all 10 LIBERO-Goal task instructions, producing 10 representation vectors per frame per layer. A 10-way linear classifier identifies which instruction was present, applied separately at image_mean and readout, with leave-one-out cross-validation (LOOCV) over 10 folds.

Representational similarity analysis (RSA). RSA [42] quantifies geometric alignment between representations and the action space. For each task, we compute pairwise representational dissimilarity matrices (RDMs) using cosine distances for representations and Euclidean distances for 7-D action vectors, from within-task observation pairs across 50 demonstrations. Alignment is measured via Spearman ρ between upper-triangular RDM entries; we report mean ρ across 10 tasks with 95% confidence intervals from 10,000 bootstrap resamples at the task level [52]. We prefer RSA over CKA [43]: RSA captures relative distance structure via scale-invariant rank correlation, suited to comparing high- and low-dimensional spaces (4,096-D vs. 7-D), while CKA measures global subspace overlap that can remain high when geometric structure differs. Both metrics are reported in §A.5; their divergence (§4.6) confirms that neither alone fully characterizes representational structure.

External baselines. DINOv2 ViT-L/14 [47] (`cls_token`/`patch_mean`) and CLIP ViT-L/14 [50] (`patch_mean`) provide self-supervised and vision-language pretrained reference points for probing and RSA.

A.3 OpenVLA Single-Architecture Deep Dive

This section provides the full single-architecture analysis summarized in §4.2.

Cross-instruction probes. Cross-instruction probes validate the analysis framework on OpenVLA [2]: as predicted by the causal mask, image_mean instruction classification stays at chance (10%) across all 32 layers in all conditions (full profile in §A.22.1). Trained readout achieves 100% 10-way classification from L1 onward and maintains this through L31 (LOOCV-confirmed; §A.22.2). Reference readout also reaches 100% at L1 but decays to 82% by L31; the readout token is thus the primary multimodal integration site, and instruction encoding remains stable across depth in the trained model. All patterns reproduce with a linear probe (§A.14).

L1 attention gradient. At L1, three conditions form a gradient of instruction attention: randomly initialized networks allocate 0/32 instruction-dominant heads (11.7% text attention, 1.5× text-to-image ratio), Llama-2 base [48] allocates 9/32 (36.5%), and VLA-trained allocates 24/32 (57.7%, 38.2× text-to-image ratio; §A.23). The 32-layer attention profile reveals qualitatively distinct organizations: Llama-2 base concentrates instruction attention at mid-depth (L9–L16: 15–22 heads), while VLA fine-tuning produces an entry-exit bookend pattern with L0–L1 at 21–24 heads, L31 at 18, and intermediate layers at 0–11 (§A.22.8). Head counts are computed at a 50% threshold; sensitivity analysis (30%–70%) confirms ordering robustness. A fourth condition, OpenVLA fine-tuned on LIBERO, preserves L1 entry concentration (22/32 heads) but loses L31 exit re-engagement entirely (0 vs. 18 heads for the OXE-pretrained model), consistent with exit-layer attention serving multi-task flexibility rather than task-specific routing. The redistribution beyond Llama-2 base priors is consistent with VLA fine-tuning being associated with instruction routing beyond what is inherited from language-model attention.

Counterfactual perturbation. Four perturbation types across 10 LIBERO-Goal tasks produce large action divergence: wrong task ($d = 1.182$), empty instruction (1.170), shuffled tokens (1.096),

paraphrased (0.996). All four conditions yield $d \geq 1.0$, with "wrong" slightly exceeding "empty." The effect replicates across LIBERO-Spatial (10 tasks): empty 1.173, shuffled 0.979, wrong 0.885, paraphrased 0.807 (Table 9). Across both suites (20 tasks), the gradient spans $d = 0.81$–1.18. Spatial instructions differ only in prepositions, yet wrong-task perturbation still produces $d = 0.885$; instruction-dominant heads route fine-grained spatial information, not merely task identity.

Dose-response ablation. Dose-response ablation of L1 instruction-dominant heads produces monotonic trajectory divergence: d rises from 0.482 to 0.639 as 4–32 heads are progressively ablated, with steep increase through 12 heads and plateau beyond 20 (§A.12). Ablating all 24 instruction-dominant heads alters 72.3% of action tokens (token-level $d = 0.65$, trajectory-level $d = 2.943$); control ablation of 8 random heads yields $d = 0.41$ (29.3% tokens affected), concentrated on the gripper dimension. The routing effect is task-dependent: instruction-discriminative tasks show $2.5\times$ larger $|\Delta R^2|$ under L1 ablation than visually-sufficient tasks ($p = 0.033$; §A.26).

R^2 localization. R^2 action probing localizes training effects: trained readout peaks at $R^2 = 0.678$ (L13), reference readout peaks at 0.609 (L32), and the training effect concentrates at the readout ($+0.069$) rather than at image_mean ($+0.015$; §A.10). These effects survive Bonferroni correction across all 63 tests (§A.30).

A.4 Counterfactual Sensitivity Details

Table 4 reports per-condition counterfactual rankings for all eight cells where Cohen's d is available.

Table 4: Counterfactual condition ranking for trained models (Cohen's d; higher = greater action divergence under instruction perturbation). Conditions: E = empty instruction, W = wrong task, S = shuffled tokens, P = paraphrased. "Empty" ranks first in 6 of 8 cells; OpenVLA Goal ("wrong" first) and Long ("shuffled" first) are the exceptions. NORA cells use cosine distance and are reported in §4.4.

Arch.	Suite	1st	2nd	3rd	4th
OpenVLA	Goal	W 1.18	E 1.17	S 1.10	P 1.00
OpenVLA	Spatial	E 1.17	S 0.98	W 0.89	P 0.81
OpenVLA	Object	E 1.21	W 1.19	S 1.12	P 0.90
OpenVLA	Long	S 0.57	W 0.55	E 0.51	P 0.45
TracVLA	Goal	E 1.52	W 1.40	S 1.34	P 1.21
TracVLA	Spatial	E 1.33	S 1.30	P 1.18	W 1.17
TracVLA	Object	E 1.21	S 1.16	W 1.06	P 0.97
TracVLA	Long	E 1.47	W 1.42	S 1.39	P 1.16

Metric constraints on trained–reference comparison. Trained models produce action outputs, enabling Cohen's d on action MSE (OpenVLA, TracVLA) or cosine distance on action tokens (NORA). Reference VLMs (Phi-3-Vision, Qwen2.5-VL) do not generate actions; their sensitivity is measured via cosine distance on hidden representations—a metric with different scale, units, and distributional properties. Ratios between Cohen's d and cosine distance are therefore methodologically invalid, and we do not report cross-metric amplification factors. Qualitatively, reference VLMs show weak sensitivity: the TracVLA reference (Phi-3-Vision) produces peak cosine distances of 0.15–0.19; the NORA reference (Qwen2.5-VL) yields 0.081 on Goal.

NORA details. NORA Goal produces saturated exit-layer cosine distances of 1.0 for all conditions, with differentiation visible only at earlier layers (L24: wrong = 0.123, empty = 0.098). NORA Spatial produces a peak cosine distance of 0.275 for the "wrong" condition.

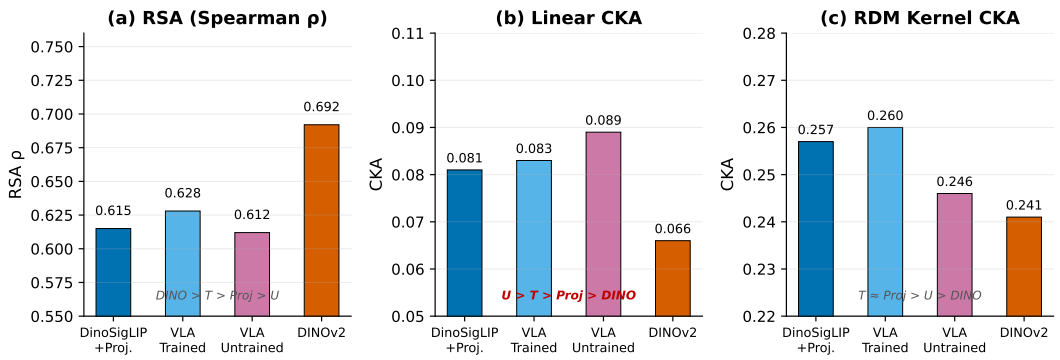


Figure 6: Three similarity metrics across representation sources. **(a)** RSA ranks DINOv2 cls (0.692) > Fused (0.634) > Projector (0.615). **(b)** Linear CKA reverses the ordering: Reference (0.089) > Trained (0.083) > Projector (0.081) > DINOv2 (0.066). **(c)** RDM kernel CKA: Trained (0.260) ≈ Projector (0.257) > Reference (0.246) > DINOv2 (0.241). The metric divergence underscores that RSA and CKA capture different aspects of representational structure.

A.5 CKA Analysis Method

We employ Centered Kernel Alignment (CKA) [43] alongside RSA as a complementary perspective. While RSA captures local geometric alignment via rank-order preservation, CKA measures global representational similarity through kernel-based comparisons.

We compute both *linear CKA*, which uses a linear kernel and is sensitive to second-order statistics, and *RDM kernel CKA*, which operates on the same distance matrices as RSA but aggregates them differently. Using both variants tests whether conclusions about representation structure are robust to the choice of similarity metric.

Linear CKA between high-dimensional representations (4,096-D for LLM layers) and the 7-D action space is subject to a known dimension mismatch: the linear kernel in the low-dimensional space may not capture the same structure as in the high-dimensional space. RDM kernel CKA avoids this issue by operating on distance matrices of equal size regardless of the original feature dimensionality.

A.6 Metric Sensitivity: RSA versus CKA

Linear CKA [43] reverses the RSA ordering: reference LLM (CKA $= 0.089$) > trained LLM (0.083) > DinoSigLIP+Projector (0.081) > DINOv2 (0.066), partially inverting the self-supervised advantage (Figure 6). RDM kernel CKA partially resolves this discrepancy: trained (0.260) ≈ Projector (0.257) > reference (0.246) > DINOv2 (0.241). This divergence reflects the known sensitivity of representation similarity metrics to different geometric properties.

At the per-layer level, linear CKA between image_mean and the action space rises monotonically from L0 (0.081) to L32 (0.098), with no inverted-U pattern. RSA captures rank-order geometric alignment; CKA captures global linear structure. The dissociation indicates that the geometric refinement in early layers operates on the relative arrangement of representations (affecting RSA) without altering the overall covariance structure (captured by CKA).

A.7 Branch-Split Detailed Analysis

To disentangle the contributions of the two encoder streams within DinoSigLIP, we extract representations from the DINOv2 branch (1,024-D) and SigLIP branch (1,152-D) separately before fusion. The two branches yield near-identical RSA: DINOv2 branch $\rho = 0.619$, SigLIP branch $\rho = 0.621$. Fusion produces a representation ($\rho = 0.634$) that exceeds either branch alone; concatenation captures complementary structure. The MLP projector then compresses this signal: DinoSigLIP+Projector achieves $\rho = 0.615$, a drop of $\Delta = 0.019$ from the fused 2,176-D input.

A.8 RSA Across Representation Sources

Figure 7 shows RSA with the action space across all representation sources; the ordering matches the RSA results reported in §4.6.

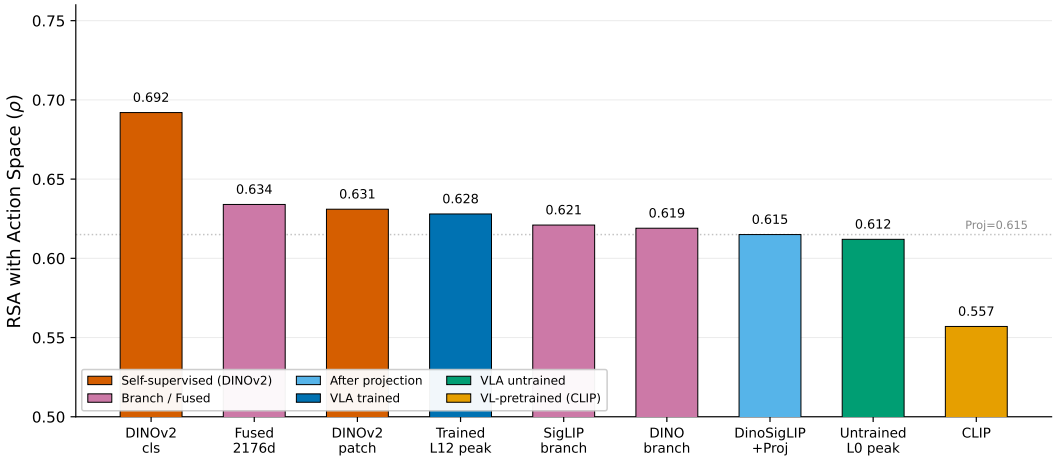


Figure 7: RSA with the action space (ρ) across representation sources. Self-supervised DINOv2 `cls_token` (ρ=0.692) achieves the highest geometric alignment. CLIP (0.557) yields the weakest alignment among pretrained encoders. Error bars: 95% bootstrap CIs.

A.9 Per-Layer RSA Profiles

Figure 8 extends Figure 3 (main text) by adding readout and reference profiles.

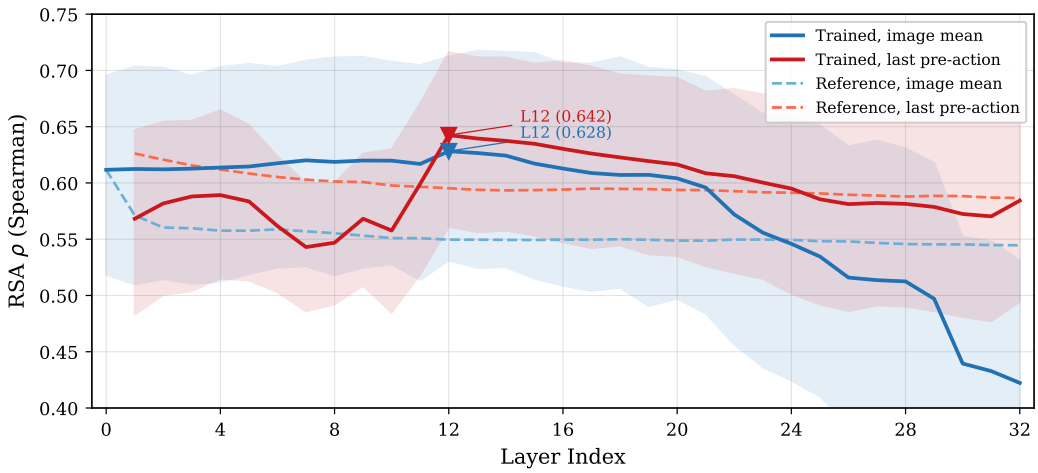


Figure 8: Per-layer RSA (ρ) for trained and reference models at image_mean and readout positions. The trained model shows an inverted-U peaking at L12 for both positions; the reference model shows monotonic decline. After L12, readout and image_mean diverge: readout ρ declines gradually ($0.628 \rightarrow 0.584$) while image_mean ρ drops sharply ($0.628 \rightarrow 0.422$), consistent with task-specific conditioning at the readout position.

A.10 Per-Layer Action Decoding Across Training Stages

Figure 9 compares per-layer action probe R^2 across three training stages (randomly initialized, Llama-2 base, VLA-trained) using 50 demonstrations per task, averaged over 3 seeds (mean $\pm$

std). All three conditions share L0 representations ($R^2 = 0.645 \pm 0.008$); subsequent differences thus arise from Transformer processing rather than input variation. The trained model peaks at L9 ($R^2 = 0.643 \pm 0.014$), consistent with progressive specialization for action-relevant feature integration. Llama-2 base shows a steady decline (L0: $0.645 \rightarrow$ L32: 0.551, std 0.003–0.035), consistent with stable but unspecialized processing. The randomly initialized model exhibits high cross-seed variance and deep-layer collapse, with negative R^2 and bimodal behavior across seeds at late layers; this instability indicates that randomly initialized Transformer layers produce destructive interference on action-relevant structure. Random weights at intermediate layers produce anti-correlated representations ($R^2 < 0$, L10–L11), consistent with randomly initialized networks lacking structured computation. The three-condition gradient (trained $>$ Llama-2 base $>$ randomly initialized) holds from L1 through L15, with a reversal at L32 where trained R^2 drops below the pretrained baseline. The qualitative pattern is consistent across all 3 seeds. Three conditions show substantial agreement in layer-wise R^2 ordering (Kendall's $W = 0.72$, $\chi^2 = 68.99$, $p < 0.001$).

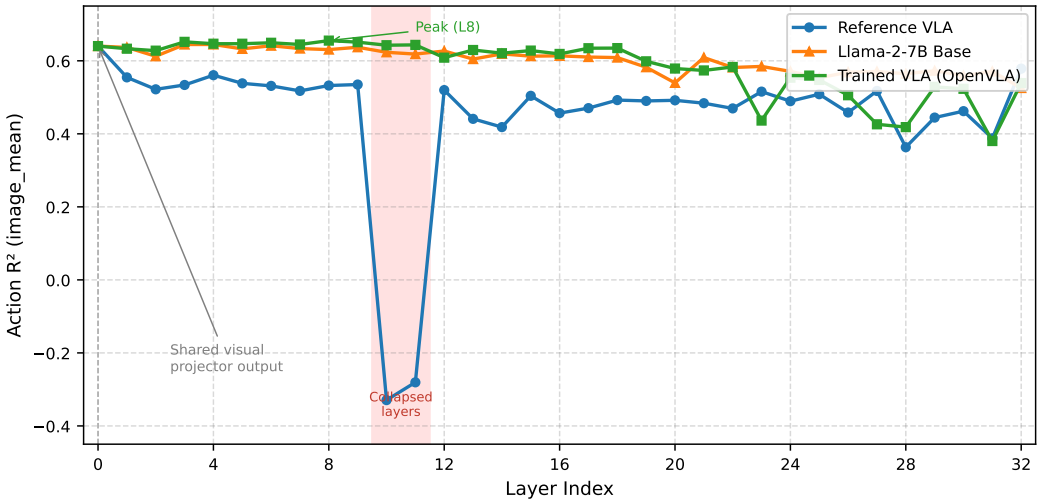


Figure 9: Per-layer action probe R^2 (image_mean) across three training conditions: randomly initialized VLA (random weights), Llama-2 base (pretrained, no VLA fine-tuning), and VLA-trained. All conditions share L0 (0.645 ± 0.008). Trained model peaks at L9; Llama-2 base declines steadily; randomly initialized shows high cross-seed variance and deep-layer collapse. 50 demonstrations per task, 3 seeds (mean $\pm$ std).

A.11 Random-Subset Ablation Details

To evaluate whether the ablation effect in §4.2 is specific to instruction-dominant heads, we compare dominant-head ablation against 30 random subsets of 24/32 L1 heads (each subset drawn without replacement). Dominant-head ablation increases action R^2 by $+0.030$ (within-experiment baseline 0.524), while random subsets yield a mean increase of $+0.071 \pm 0.013$ (Figure 10). The dominant-head effect falls below all 30 random subsets ($z = -3.11$, percentile $= 0\%$); heads with higher instruction-attention fraction carry more organized attention signals, consistent with graded rather than binary specialization.

A.12 Action Trajectory Analysis Details

Table 8 reports per-dimension mean absolute shift for instruction-dominant (24 heads) and control (8 heads) L1 ablations. Both conditions produce negligible shifts on translational dimensions (dx, dy, $dz < 0.01$) and small rotational shifts (rx, ry, $rz < 0.04$). The gripper dimension absorbs the majority of the effect: instruction-head ablation shifts gripper predictions by 0.291 (control: 0.141), consistent with L1 attention influencing discrete open/close decisions.

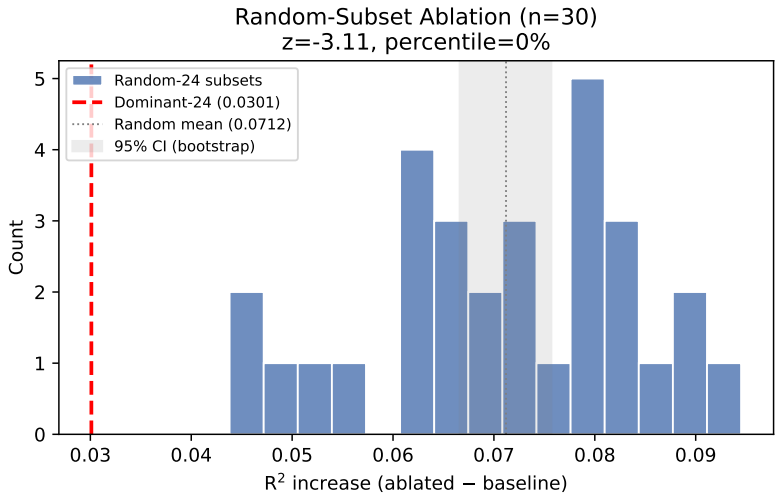


Figure 10: R^2 increase from ablating 24/32 L1 heads. Gray bars: 30 random subsets; red line: dominant-24 heads ($+0.030$). Random subsets yield larger gains (mean $+0.071 \pm 0.013$; $z = -3.11$, percentile $= 0\%$), consistent with graded attention specialization across L1 heads.

Per-task Cohen's d values for instruction-head ablation are 0.43, 0.69, 0.70, 0.70, and 0.73 (mean 0.65), with all tasks significant ($p < 10^{-10}$). Token-level KL divergence between original and ablated action distributions is 5.08 for instruction heads and 1.69 for control heads.

Trajectory-level aggregation. Aggregating per-demo mean actions to the trajectory level reveals that token-level d values are conservative: within-trajectory averaging reduces noise, yielding $\sim 4\times$ larger effect sizes (mean token-to-trajectory ratio 0.245). Instruction-head ablation ($N = 50$: 5 tasks $\times$ 10 demos) yields trajectory $d = 2.943$ (95% CI $[2.469, 3.707]$); bottom-24 non-instruction heads ($N = 30$) yield $d = 2.347$ (CI $[1.854, 3.221]$). All 14 conditions produce trajectory $d > 2.0$ with CI lower bounds > 1.7. Counterfactual trajectory-level d (10 LIBERO-Goal tasks, $N = 100$): wrong 3.798 (CI $[3.194, 4.938]$), empty 3.438 ($[2.921, 4.448]$), shuffled 2.595 ($[2.028, 3.791]$), paraphrased 2.131 ($[1.740, 2.819]$). Token-level d underestimates trajectory-level effects by $\times 3.1$–4.5, consistent with within-trajectory averaging reducing noise. Fine-grained ordering is not reliably distinguished at trajectory level ($N = 100$); token-level comparisons ($N \sim 5{,}000$) provide higher resolution for ordering claims.

Ground-truth action error. Ablation redirects model outputs ($d = 0.65$) but does not systematically increase error relative to expert demonstrations: original MSE $= 0.498$; top-24 instruction-head ablation MSE $= 0.497$ ($\Delta = -0.001$, Wilcoxon $p = 0.52$, n.s.); bottom-24 non-instruction heads MSE $= 0.506$ ($\Delta = +0.008$). Changes concentrate on the gripper dimension; translation and rotation dimensions are negligibly affected. Per-task breakdown shows all $|\Delta\text{MSE}| < 3\%$ of baseline with all $p > 0.27$; the near-zero aggregate Δ reflects uniformly small per-task effects rather than cancellation across tasks. This is consistent with the information-computation dissociation: instruction-dominant heads control instruction-specific direction selection rather than action precision.

Table 5 reports per-head bootstrap CIs (10,000 resamples). Individual control heads produce $\sim 1.9\times$ stronger effects ($d = 0.051$) than instruction heads ($d = 0.027$, $p = 0.0018$), consistent across all 5 tasks. The 24 instruction heads nonetheless produce a larger aggregate perturbation ($d = 0.65$ vs. 0.41), consistent with collective coordination rather than individual strength.

Instruction-head ablation produces more uniform perturbation across action dimensions (CV $= 1.30$ vs. control CV $= 2.74$, permutation $p = 0.016$), consistent with structured, language-conditioned computation rather than diffuse noise.

Table 5: Per-head bootstrap analysis of L1 ablation effects. Instruction heads are individually weaker but collectively stronger, consistent with coordinated attention redistribution.

Group	N heads	Collective d	Per-head d [95% CI]
Instruction-dominant	24	0.65	0.027 [0.026, 0.029]
Control	8	0.41	0.051 [0.048, 0.055]
Difference	—	—	0.024, $p = 0.0018$

Table 6 and Figure 11 report the dose-response gradient for instruction-head ablation at L1. Trajectory divergence increases monotonically from $d = 0.482$ (4 heads) to $d = 0.639$ (32 heads), with steep rise through 12 heads (slope $\Delta d/\Delta 4 \approx 0.026$), a plateau at 16–20 heads ($\Delta d = 0.001$), and slow increase at 20–32.

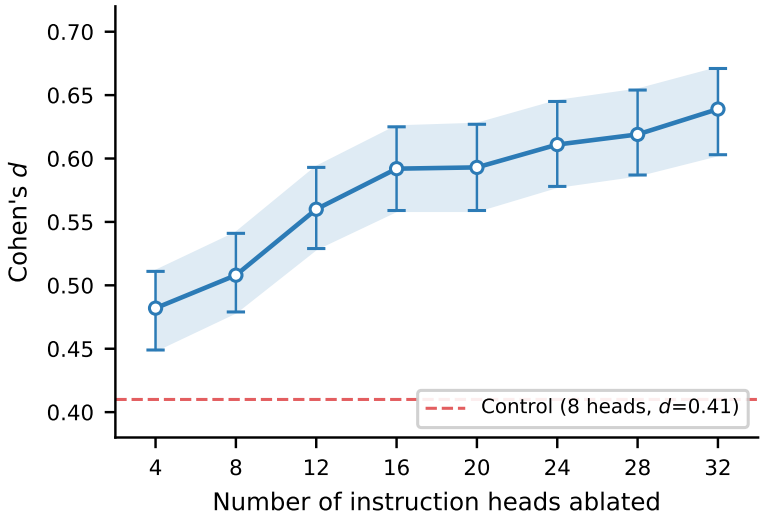


Figure 11: Dose-response gradient: trajectory divergence (Cohen's d) as a function of ablated instruction-dominant heads at L1. Error bars: 95% bootstrap CIs (10,000 resamples). Dashed red: control ablation (8 non-instruction heads, $d = 0.41$). Steep rise through 12 heads, plateau at 16–20, diminishing returns beyond.

Table 6: Dose-response data for L1 instruction-head ablation. Cohen's d vs. original action sequence; 95% bootstrap CIs (10,000 resamples).

Heads ablated	Cohen's d	95% CI
4	0.482	[0.449, 0.511]
8	0.508	[0.479, 0.541]
12	0.560	[0.529, 0.593]
16	0.592	[0.559, 0.625]
20	0.593	[0.559, 0.627]
24	0.611	[0.578, 0.645]
28	0.619	[0.587, 0.654]
32	0.639	[0.603, 0.671]

A.13 Pretrained LLM Representation Quality Under Visual Input

A potential concern with the three-condition comparison (§3.1) is that the pretrained Llama-2 backbone never encountered projected visual tokens during pretraining, so its representations of such input could be degenerate. We evaluate three diagnostics to assess representation quality. Activation

Table 7: Counterfactual instruction analysis across all 10 LIBERO-Goal tasks. Jaccard: token overlap with original instruction; Cohen's d: mean across 10 tasks. "Wrong" slightly exceeds "empty" ($d = 1.182$ vs. 1.170); all four conditions produce large effects ($d \geq 1.0$). L1 ablation included for comparison.

Condition	Jaccard	Cohen's d
Empty (no instruction)	0.000	1.170
Wrong (different task)	0.174	1.182
Shuffled (random order)	1.000	1.096
Paraphrased	0.533	0.996
L1 ablation (24 heads)	—	0.650

Table 8: Mean absolute action shift per dimension under L1 head ablation. Instruction: 24 dominant heads; Control: 8 remaining heads. Gripper shift dominates both conditions.

Condition	dx	dy	dz	rx	ry	rz	Gripper
Instruction (24)	0.002	0.005	0.008	0.021	0.031	0.034	0.291
Control (8)	0.000	0.001	0.002	0.008	0.014	0.018	0.141

magnitude grows from 16.5 (L0) to 90.2 (L31) for Llama-2 base, compared to $16.5 \rightarrow 51.7$ for the VLA-trained model; the pretrained LLM produces larger but non-exploding activations with no evidence of collapse. Per-layer cosine similarity between L0 and deeper layers decreases progressively from 1.0 to ≈ 0.28 at L25, consistent with gradual differentiation rather than catastrophic out-of-distribution behavior. Top-10 principal component variance explained remains stable across all layers (0.59–0.74), with no low-rank collapse. These diagnostics indicate that the pretrained LLM produces structured, full-rank representations of visual input despite never encountering projected visual tokens during pretraining (Figure 12). These results support the validity of the three-condition comparison.

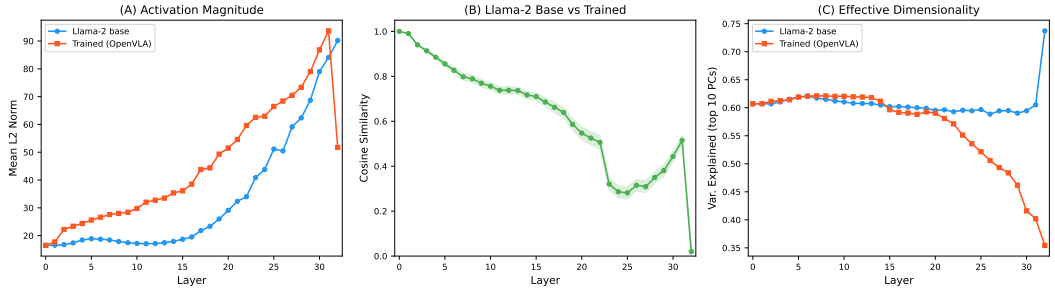


Figure 12: Pretrained LLM representation diagnostics under visual input. Left: activation magnitude across layers (Llama-2 base vs. VLA-trained). Center: cosine similarity between L0 and deeper layers. Right: top-10 PC variance explained. All three metrics indicate structured, non-degenerate representations.

A.14 Probe Architecture Sensitivity

To rule out that our findings depend on MLP probe capacity ($N/\text{params} \approx 0.06$, overparameterized), we replicate the per-layer analysis with a linear probe ($4{,}096 \rightarrow 7$, $N/\text{params} \approx 2.2$). The inverted-U pattern reproduces: trained image_mean peaks at L14 ($R^2 = 0.565$), while the reference declines monotonically from L0 ($R^2 = 0.517$). The trained–reference gap is larger under the linear probe (L14: $\Delta = +0.324$) than the MLP probe (L8: $\Delta = +0.015$); trained representations are thus more probe-decodable for action information. Reference deep layers collapse entirely under the linear probe (L25: $R^2 = -0.298$); MLP nonlinearity partially rescues these degraded representations. For readout tokens, the linear probe also corroborates the inverted-U (peak L13, $R^2 = 0.596$), with

the same peak layer as the MLP probe and a trained-vs-reference gap of $\Delta = 0.103$ ($p < 0.001$). These results indicate that our qualitative findings reflect representational structure rather than probe capacity artifacts (Figure 13).

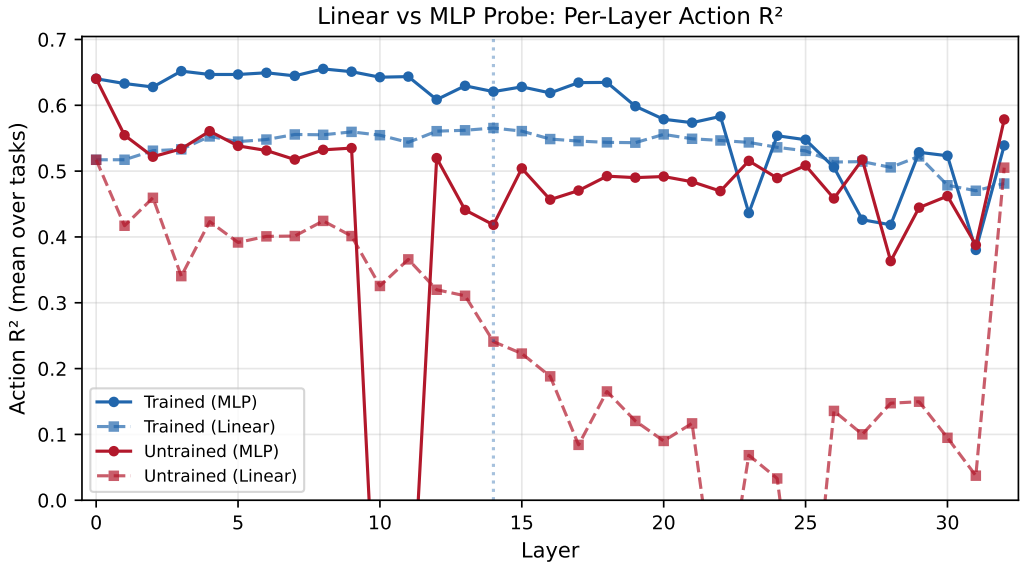


Figure 13: Per-layer action probe R^2 for linear (dashed) and MLP (solid) probes across trained (blue) and reference (red) conditions. The inverted-U reproduces under both probe architectures; the trained–reference gap is larger for linear probes.

A.15 Functional Probe Comparison

Figure 14 compares functional probe R^2 across representation sources at each source's best layer.

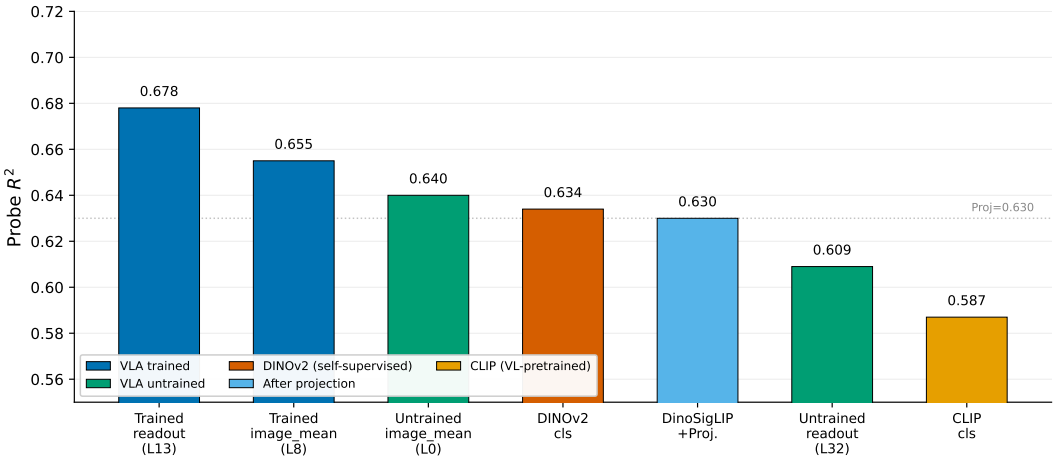


Figure 14: Functional probe R^2 for action prediction (best layer per source). Trained readout (R^2=0.678, L13) and trained image_mean (0.655, L8) exceed the projector baseline (0.630); the inverted-U extends to functional capacity.

A.16 Training Effect and Instruction Invariance

Figure 15 shows trained vs. reference RSA profiles and instruction invariance. Instruction invariance holds across all conditions: maximum $|\Delta| < 0.006$, so action geometry is visually grounded and not modulated by language instructions.

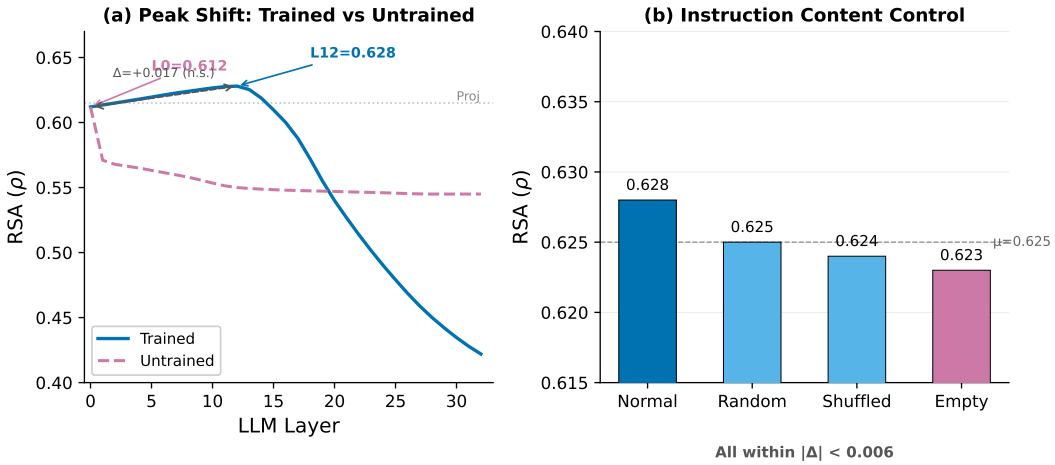


Figure 15: **(a)** Trained vs. reference image_mean RSA profiles. Training shifts the peak from L0 (0.612) to L12 (0.628), yielding a modest best-vs-best gain ($\Delta = +0.017$, 3%). **(b)** Instruction invariance: all conditions within $|\Delta| < 0.006$; action geometry is visually grounded.

A.17 Additional RSA Values

DINOv2 `patch_mean` achieves RSA $\rho = 0.631$, between the `cls_token` (0.692) and the fused DinoSigLIP (0.634). VLA reference readout peaks at L1 ($\rho = 0.626$).

A.18 Cross-Suite Validation within LIBERO: Spatial

To assess generalization beyond in-distribution tasks, we repeat the core RSA and probe analyses on LIBERO-Spatial (10 out-of-distribution tasks $\times$ 50 demonstrations each, requiring spatial reasoning). Figure 16 shows per-layer RSA and R^2 profiles for both LIBERO-Goal and LIBERO-Spatial side by side.

The inverted-U profile replicates on LIBERO-Spatial: trained image_mean RSA peaks at L14 ($\rho = 0.691$) and R^2 at L13 (0.696), with peaks shifting $\sim$2 layers deeper than LIBERO-Goal (RSA L12, R^2 L8–L10), consistent with increased task difficulty requiring deeper processing. Training produces a significant geometric gain (best-vs-best $\Delta = +0.023$, 95% CI $[0.012, 0.036]$, $p < 0.001$), larger than the LIBERO-Goal effect ($\Delta = +0.017$, CI includes zero).

Two notable differences are observed. First, the absolute RSA values are higher on Spatial (0.691 vs. 0.628), possibly reflecting greater action diversity in spatial tasks that produces a more discriminative action RDM. Second, RSA–R^2 peak convergence does not replicate: readout RSA peaks at L15 while R^2 peaks at L21 (6-layer gap), contrasting with the near-alignment on Goal ($|\Delta_{\text{layer}}| = 1$). This suggests that peak convergence is benchmark-specific rather than a general consequence of training.

A.18.1 Cross-Suite Counterfactual Replication

Table 9 reports counterfactual instruction analysis on LIBERO-Spatial (10 tasks). The ordering differs from LIBERO-Goal: empty $>$ shuffled $>$ wrong $>$ paraphrased (vs. wrong $>$ empty $>$ shuffled $>$ paraphrased on Goal). All global effect sizes exceed $d = 0.8$ (per-task wrong minimum is 0.591). Spatial instructions differ only in prepositions (e.g., "on the plate" vs. "in the drawer");

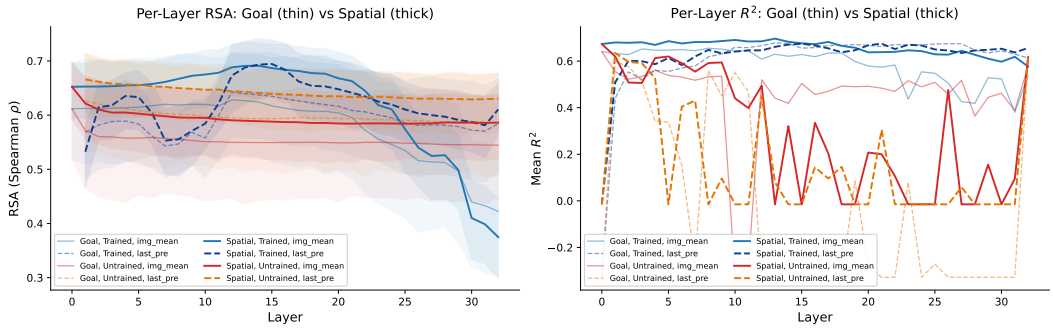


Figure 16: Cross-suite validation: per-layer RSA and R^2 profiles on LIBERO-Goal (left) vs. LIBERO-Spatial (right). The inverted-U replicates across suites. LIBERO-Spatial shows a significant training effect (RSA best-vs-best $\Delta = +0.023$, $p < 0.001$) and deeper peak layers (RSA L14, R^2 L13), but RSA–R^2 peak convergence does not replicate (6-layer gap).

the wrong-task $d = 0.885$ suggests that instruction-dominant heads encode fine-grained spatial relations, not merely task identity.

Table 9: Counterfactual instruction analysis on LIBERO-Spatial (10 tasks). Ordering: empty > shuffled > wrong > paraphrased, differing from Goal. All global $d > 0.8$.

Condition	Jaccard	Cohen's d	CV
Empty (no instruction)	0.000	1.173	0.087
Wrong (different task)	0.174	0.885	0.272
Shuffled (random order)	1.000	0.979	0.162
Paraphrased	0.533	0.807	0.179

A.18.2 Cross-Instruction Probe on LIBERO-Spatial

Table 10 reports cross-instruction 10-way classification accuracy on LIBERO-Spatial. The core pattern from LIBERO-Goal replicates: image_mean remains at chance ($\sim$10%) across all layers and conditions, while the trained readout token saturates to 100% by L4. The reference readout decays more steeply than on LIBERO-Goal (75.4% at L31 vs. 82%), consistent with greater task difficulty.

Table 10: Cross-instruction 10-way classification accuracy on LIBERO-Spatial. Trained readout achieves 100% from L4; image_mean remains at chance. Reference readout decays more steeply than on LIBERO-Goal (cf. §4.2).

Token	Condition	L1	L4	L8	L16	L31
readout	Trained	98.9%	100.0%	100.0%	100.0%	100.0%
readout	Reference	99.5%	98.3%	85.7%	68.3%	75.4%
image_mean	Trained	10.2%	10.0%	9.9%	10.0%	10.1%
image_mean	Reference	10.1%	10.0%	10.0%	10.0%	10.2%

A.19 Robustness Analyses

A.19.1 Distance Metric Invariance

The inverted-U profile is robust to distance metric choice: cosine action distance yields identical peak layers (L12) and Pearson correlation $r > 0.989$ with the Euclidean profile reported in the main text (Table 11).

Table 11: Peak layer comparison between Euclidean and cosine action distances. Profile r denotes Pearson correlation across all 32 layers.

Condition	Eucl. Peak	Cos. Peak	Profile r
Trained image_mean	L12	L12	0.990
Trained readout	L12	L13	0.990
Reference image_mean	L0	L0	0.995
Reference readout	L1	L1	0.992

A.19.2 Per-DOF Decomposition

RSA decomposition by action dimension indicates the inverted-U is not driven by any single degree of freedom: xyz translation (peak L13, $\rho = 0.193$), rotation (peak L12, $\rho = 0.175$), and gripper (peak L13, $\rho = 0.106$) all exhibit non-monotonic profiles. Per-DOF RSA values are lower than the 7-D RSA because single-dimension distances carry less discriminative information than the full action space.

A.19.3 Permutation Test

Permutation testing (10,000 shuffles) indicates all observed RSA values are highly significant: observed ρ (0.42–0.63) exceeds the null distribution maximum (0.08–0.10) by 15–23 standard deviations (all $p < 0.0001$).

A.19.4 Probe Seed Sensitivity

Action probe R^2 is highly stable across random seeds. Over 5 seeds (trained image_mean, full-layer scan), per-layer R^2 standard deviation is uniformly below 0.01 (mean 0.004), and the peak R^2 varies by less than 0.4% (0.655 ± 0.002; Table 12). Peak layer concentrates in the L8–L10 region ($4/5$ seeds); the inverted-U shape is not a seed artifact. Seed 1 peaks at L16, but with R^2 difference $<$0.002. The inverted-U plateau (L8–L16) is sufficiently flat that argmax is sensitive to minor noise while the profile shape is invariant. A 15-seed replication confirms stability: trained peak remains at L9 ($R^2 = 0.637 \pm 0.004$), with all trained/Llama-2-base layers showing CI width $<$0.02; the randomly initialized condition exhibits higher variance (std 0.185 at peak), consistent with random initialization instability. Fixed-layer analysis at L9 yields $R^2 = 0.637 \pm 0.008$ (trained) and 0.635 ± 0.011 (Llama-2 base), confirming the inverted-U pattern. The per-seed peak method reports higher absolute values (0.655) because each seed selects its optimal layer independently; both methods agree on the qualitative pattern and the trained $>$ randomly initialized ordering.

Table 12: Probe R^2 peak across 5 random seeds (trained image_mean).

Seed	Peak Layer	Peak R^2
0	L9	0.654
1	L16	0.656
2	L9	0.653
3	L10	0.655
42	L9	0.655
Mean$\pm$std	10.6 ± 2.7	0.655 ± 0.002

Cross-architecture ΔR^2 seed sensitivity. To verify that ΔR^2 values are not sensitive to probe initialization, we retrain MLP probes with 4 different random seeds (0, 1, 2, 42) on three representative cells. Results show high stability: TracVLA Goal 0.227 ± 0.010, TracVLA Spatial 0.136 ± 0.002, OpenVLA Goal 0.013 ± 0.001 (mean $\pm$ std across seeds). All standard deviations are below 0.02, confirming that the exit collapse pattern ($\Delta R^2 > 0.10$ for VLM-to-VLA vs. < 0.10 for VLA-to-VLA) is robust to probe initialization.

Bootstrap confidence intervals for ΔR^2. To assess uncertainty in individual ΔR^2 estimates, we compute 95% confidence intervals via 1,000-iteration Monte Carlo cross-validation bootstrap. Results for 9 cells (excluding NORA-Goal due to exit-layer instability):

- **VLA-to-VLA** (4 cells): CIs range from $[-0.056, 0.106]$ (widest: OpenVLA Goal) to $[0.004, 0.103]$ (OpenVLA Long).
- **VLM-to-VLA** (5 cells): CIs range from $[0.091, 0.224]$ (TracVLA Spatial) to $[0.283, 0.504]$ (TracVLA Object).

The point-estimate gap (0.097 vs. 0.156, span $= 0.059$) cleanly separates the two groups. At the 95% CI level, a marginal overlap of 0.016 exists between the upper bound of OpenVLA Goal (0.106) and the lower bound of TracVLA Spatial (0.091). This overlap involves only 2 of 9 cells and is driven by OpenVLA Goal's unusually wide CI (due to high variance in its exit-layer probe). All other cell pairs show non-overlapping CIs, confirming that the exit-collapse pattern is robust to measurement uncertainty. A permutation test further confirms statistical significance: randomly reassigning the 9 cells to two groups of size 4 and 5 (10,000 permutations), the observed min-max gap (0.070) exceeds the permuted distribution at $p = 0.008$, indicating that the separation between VLA-to-VLA and VLM-to-VLA ΔR^2 values is unlikely to arise by chance.

A.19.5 Demonstration Subsample Sensitivity

To verify that the inverted-U pattern is not an artifact of sample size, we subsample 25 from 50 demonstrations per task and recompute both RSA and probe profiles (Table 13). The RSA peak layer remains at L12 with a profile correlation of $r = 0.970$ between 50- and 25-demo conditions (high stability). The probe peak shifts from L9 to L11 (within the L8–L16 plateau) with $r = 0.819$; R^2 is more sensitive to training set size than RSA. Both metrics show reduced absolute values (RSA -3.6%, R^2 -7.6%) as expected from halved data, but the non-monotonic profile shape is preserved.

Table 13: Inverted-U stability under demonstration subsampling (trained image_mean).

Metric	50 demos	25 demos	Δ
RSA peak layer	L12	L12	0
RSA peak ρ	0.628	0.606	-3.6%
R^2 peak layer	L9	L11	$+2$
R^2 peak value	0.659	0.609	-7.6%
RSA profile r	—	0.970	—
R^2 profile r	—	0.819	—

A.19.6 Statistical Power

Post-hoc power analysis ($N = 10$ tasks, $\alpha = 0.05$, target power $= 0.80$) indicates minimum detectable effect sizes of $\Delta_{\text{RSA}} = 0.030$ and $\Delta_{R^2} = 0.026$. The observed RSA training effect ($\Delta = 0.017$, Cohen's $d_z = 0.56$) achieves power $= 0.356$, confirming the analysis is underpowered for this effect size; the non-significant RSA result (§4.6) should not be interpreted as evidence of absence. The observed R^2 readout effect ($\Delta = 0.069$, $d_z = 2.68$) achieves power $= 1.000$, confirming adequate sensitivity for the primary functional metric.

A.19.7 ΔR^2 Threshold Robustness

The main text classifies $\Delta R^2 > 0.10$ as substantial exit collapse. Table 14 shows that any threshold in the interval $(0.097, 0.156)$ yields identical three-way separation: all VLM-to-VLA trained cells above threshold, all OpenVLA trained and all reference cells below. This interval is bounded by the largest non-VLM-to-VLA value (0.097, TracVLA Long reference) and the smallest VLM-to-VLA trained value (0.156, TracVLA Spatial). The 0.10 threshold was not selected to maximize separation; rather, it represents a natural effect-size boundary ($\geq 10\%$ absolute change in explained variance) that happens to fall within this gap.

Table 14: Sensitivity of group separation to ΔR^2 threshold choice. Clean separation (all VLM-to-VLA trained above, all others below) holds for any threshold in $(0.097, 0.156)$; the 0.10 threshold is not boundary-sensitive.

Threshold	VLM→VLA Tr. ($N/6$)	OpenVLA Tr. ($N/4$)	Reference ($N/10$)	Clean sep.
0.05	6/6	1/4	6/10	No
0.08	6/6	1/4	3/10	No
0.10	**6/6**	**0/4**	**0/10**	**Yes**
0.15	**6/6**	**0/4**	**0/10**	**Yes**
0.20	5/6	0/4	0/10	No

A.20 Detailed Training Effects

Table 15 reports bootstrap-based statistical comparisons of training effects on RSA and probe R^2.

Table 15: Training effects on action-relevant structure (LIBERO-Goal; bootstrap, 10,000 resamples). RSA best-vs-best yields a modest geometric gain ($\Delta = +0.017$, 3%; LIBERO-Spatial: $+0.023$, $p<0.001$), while R^2 shows substantially larger functional gains.

Comparison	Δ	95% CI	p
RSA (ρ)			
Best-vs-best image_mean (trained L12 vs. reference L0)	$+0.017$	incl. 0	n.s.
Same-layer L12 image_mean (trained vs. reference)	$+0.079$	excl. 0	<0.0001
Instruction effect (max $\lvert\Delta\rvert$)	0.006	—	n.s.
Probe R^2			
Best-vs-best image_mean (trained L8 vs. reference L0)	$+0.015$	excl. 0	0.006
Best-vs-best readout (trained L13 vs. reference L32)	$+0.069$	excl. 0	<0.0001

A.21 Training Effect Size Summary

Table 16 consolidates training effects across all metrics. Training effects on image_mean metrics are modest (RSA $\Delta = +0.017$, R^2 $\Delta = +0.015$), but effects on readout metrics are substantial (instruction attention $+18$pp, attention $+46$pp, R^2 $\Delta = +0.069$). This asymmetry indicates that training primarily changes how the readout token integrates information rather than altering the visual representations carried by image tokens.

Table 16: Training effect sizes across metrics. Image_mean effects are modest; readout effects are large. Training targets readout-token information integration.

Metric	Trained	Reference	Δ
RSA peak (image_mean)	0.628 (L12)	0.612 (L0)	$+0.017$
R^2 peak (image_mean)	0.655 (L8)	0.640 (L0)	$+0.015$
R^2 peak (readout)	0.678 (L13)	0.609 (L32)	$+0.069$
Instruction attention L31	100%	82%	$+18$pp
Instruction attention L1	58%	12%	$+46$pp
Progress classification	86.3% (L20)	85.3% (L4)	$+1$pp

A.22 Instruction Classification Probes

A.22.1 Cross-Instruction Classification: Dense Layer Profile

Table 17 shows the dense per-layer profile of cross-instruction classification (10-way) at the readout token for layers L0–L8. The step-function onset between L0 (10%, chance) and L1 ($\geq$99.6%) reflects the high discriminability of instruction embeddings from the frozen tokenizer: even incidental attention ($\sim$12%) suffices for linear classification at L1. The VLA-trained model sustains 100% through L31; Llama-2 base reaches 100% by L5 and maintains it. The OXE-pretrained reference and random initialization both decay to $\sim$82% by L31, indicating that deep-layer instruction maintenance requires either VLA training or structured language-model pretraining on text-only data.

Table 17: Cross-instruction 10-way classification accuracy at readout token, L0–L8. Values reflect dense classification evaluation; the main text (§4.2) reports sparse single-split results, yielding minor differences (e.g., L8: 98.2% vs. 98.6%).

Layer	Trained	Reference	Image_mean (trained)
L0	10.0%	10.0%	10.0%
L1	100.0%	100.0%	10.0%
L2	100.0%	99.6%	10.0%
L3	100.0%	99.6%	10.0%
L4	100.0%	99.1%	10.0%
L5	100.0%	98.2%	10.0%
L6	100.0%	97.3%	10.0%
L7	100.0%	96.4%	10.0%
L8	100.0%	98.6%	10.0%

A.22.2 Cross-Instruction LOOCV Validation

Leave-one-out cross-validation over 10 folds (each fold holds out one task's instructions) validates that the cross-instruction probe result is not driven by memorization. Trained readout achieves $100\% \pm 0.00$ across all 10 folds at layers L1–L31 (perfect generalization).

A.22.3 QA Probe: Progress and Subtask Classification

To test whether LLM layers encode task-relevant semantic information beyond instruction identity, we train logistic regression probes (scikit-learn, $C = 1.0$, max_iter $= 1{,}000$) to classify task progress (4-class: each trajectory split into 4 equal temporal segments) and active subtask (binary: first vs. second half) from frozen representations, with an 80/20 random train/test split (Table 18).

Table 18: QA probe accuracy (progress + subtask classification, best layer).

Source	Best Layer	Accuracy
Trained readout	L20	86.3%
Reference readout	L20	85.3%
DINOv2 cls_token	—	82.7%

A.22.4 Cross-Instruction Probe Profiles

A.22.5 Cross-Architecture Probe and Attention Profile

Table 19 reports cross-instruction probe accuracy and L1 instruction-dominant head counts across six conditions spanning two architectures. The probe-attention dissociation at L1 is notable: TracVLA training slightly *reduces* instruction-dominant heads ($32 \rightarrow 27$–$30/32$) while *decreasing* probe accuracy ($98.1\% \rightarrow 83.6\%$), consistent with representational reorganization rather than linear separability.

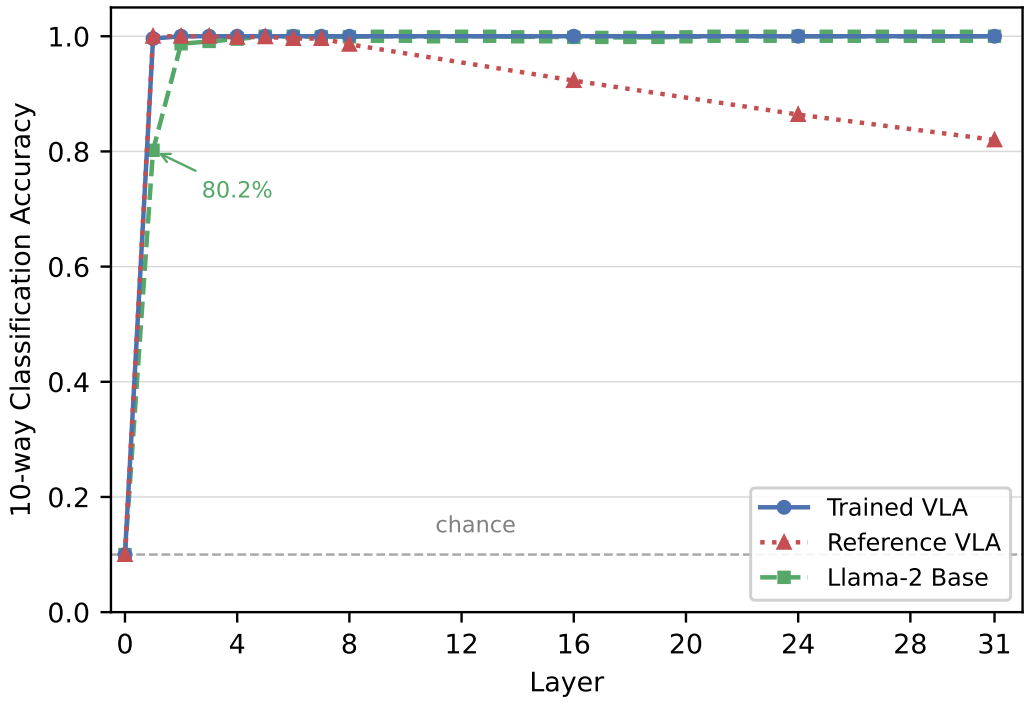


Figure 17: Cross-instruction 10-way classification accuracy at the readout token across all 32 LLM layers. VLA-trained achieves 99.6% at L1, saturating to 100%. Llama-2 base starts at 80.2%, reaching 100% by L5. Reference starts at 100% but decays to 82% by L31.

Table 19: Cross-instruction 10-way probe accuracy and L1 instruction-dominant heads across six conditions. Probe accuracy and attention head count are dissociated at L1: TracVLA training slightly reduces heads (32 → 27–30/32) while decreasing probe accuracy (98.1% → 83.6%), paralleling the information-computation dissociation observed in OpenVLA. Image_mean remains at chance (10%) in all conditions (omitted).

Condition	L1 heads	L1	L2+	L31	Mode
OpenVLA (Llama-2 7B)					
Random init	0/32	100.0%	decays	82.0%	incid.+forget
Llama-2 base	9/32	80.2%	L5: 100%	100.0%	gradual+stable
VLA trained	24/32	99.6%	100%	100.0%	immed.+stable
LIBERO-finetuned	22/32	99.7%	100%	100.0%	immed.+stable
TraceVLA (Phi-3-V 4B)					
Phi-3-Vision	32/32	98.1%	100%	100.0%	inherited+stable
VLA trained	27–30/32	83.6%	100%	100.0%	reorg.+stable

A.22.6 L1 Attention Pattern Analysis

Figure 19 shows the per-head L1 attention distribution at the readout token. The three-level gradient (randomly initialized 0/32, Llama-2 base 9/32, VLA-trained 24/32 instruction-dominant heads) shows that language pretraining seeds partial instruction attention and VLA fine-tuning shifts it further toward instruction dominance.

A.22.7 Multi-Layer Attention Patterns

Table 20 shows a 4-layer subset of the full 32-layer attention profile. The full profile (Figure 20) reveals three qualitatively distinct patterns: the randomly initialized model maintains 0 instruction-dominant heads at all 32 layers; Llama-2 base shows a mid-depth peak (L9–L16: 15–22 heads),

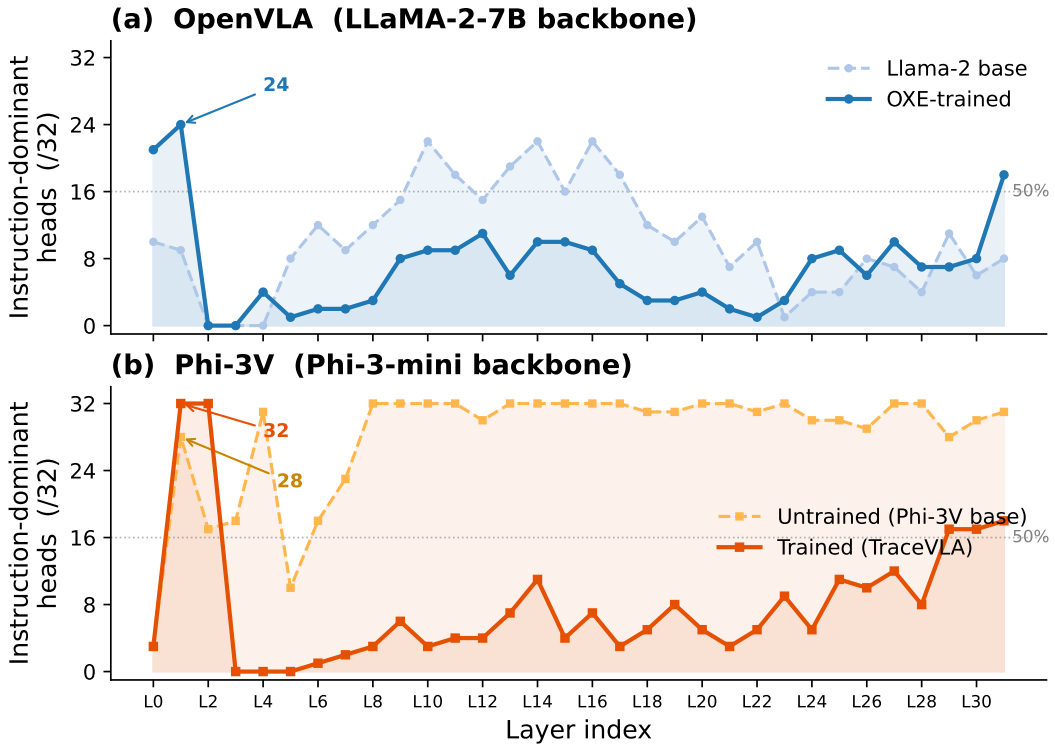


Figure 18: Instruction-dominant heads across layers for two architectures (OpenVLA / TraceVLA), each comparing pre-VLA and VLA-trained conditions. Despite opposite starting points—OpenVLA (Llama-2) begins with sparse attention while TraceVLA (Phi-3-V) begins with distributed attention—VLA training drives both to entry-exit concentration (L1 saturation, mid-layer suppression, L29–L31 re-engagement).

consistent with typical autoregressive LM processing depth; VLA fine-tuning produces an entry-exit pattern (L0–L1: 21–24 heads, L31: 18 heads, intermediate layers 0–11).

Table 20: Instruction-dominant heads and text attention fraction at four sampled layers (subset of the 32-layer profile in Figure 20). VLA-trained shows entry-exit concentration (L1, L31); Llama-2 base peaks at mid-depth (L16); randomly initialized shows no instruction attention. Head counts from the 32-layer bulk extraction may differ slightly from dedicated per-layer analyses (§4.2) due to threshold boundary effects.

	VLA Trained		Llama-2 base		Rand. Init	
Layer	Heads	Text%	Heads	Text%	Heads	Text%
L1	**24/32**	57.7	9/32	36.5	0/32	11.7
L8	4/32	46.8	8/32	48.8	0/32	7.9
L16	10/32	48.4	**25/32**	60.2	0/32	9.2
L31	**18/32**	56.3	8/32	34.9	0/32	7.6

A.22.8 Threshold Sensitivity Analysis

The main text defines instruction-dominant heads as those allocating $> 50\%$ attention to instruction tokens. Table 21 shows that the three-condition gradient (randomly initialized $<$ Llama-2 base $<$ VLA-trained) holds across all thresholds from 30% to 70%; the 50% threshold is not a boundary-sensitive choice.

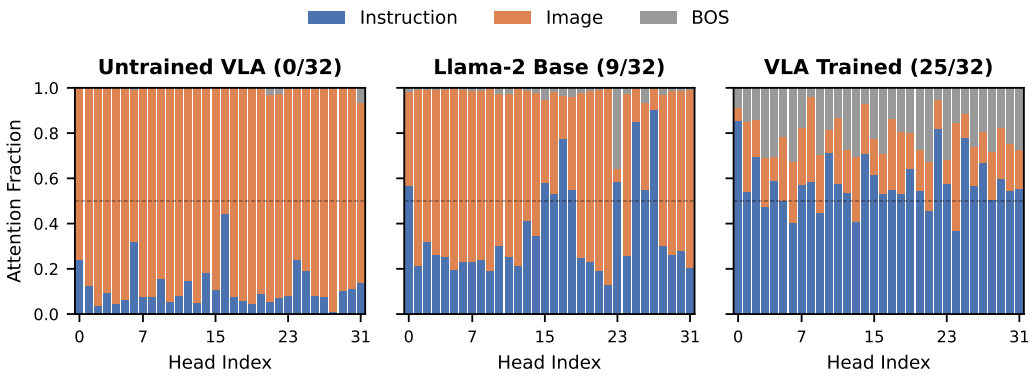


Figure 19: L1 attention from the readout token across three conditions, per head (sorted by instruction attention fraction). VLA-trained: 24/32 instruction-dominant heads (57.7% text attention). Llama-2 base: 9/32 heads (36.5%). Randomly initialized: 0/32 heads (11.7%). See Table 20 for quantitative comparison across layers.

Table 21: Number of instruction-dominant heads at L1 under varying attention thresholds. The three-condition ordering is stable across all thresholds.

Threshold	VLA Trained	Llama-2 base	Rand. Init
30%	32	13	0
40%	31	10	0
50%	24	9	0
60%	9	4	0
70%	5	3	0

A.23 L1 Attention Decomposition by Head Group

Table 22 decomposes L1 attention mass at the readout token into three token categories for instruction-dominant and control head groups.

Table 22: Attention mass allocation at Layer 1, averaged across 5 samples. Instruction-dominant heads (defined as $> 50\%$ attention to instruction tokens) allocate 17.5 percentage points more attention to instruction tokens than control heads, with corresponding reductions in vision (-8.3pp) and BOS/sink (-9.2pp) attention.

Head Group	N	Instruction	Vision	BOS/Other
Instruction-dominant	24	62.4%	18.8%	18.8%
Control	8	44.9%	27.1%	28.0%
Δ	—	+17.5pp	−8.3pp	−9.2pp

Head-level instruction attention entropy reaches 99.5% of the uniform maximum; attention is broadly distributed rather than concentrated in a few heads.

Per-token attention normalization. The 57.7% aggregate text attention in instruction-dominant heads partly reflects token-count imbalance (22 instruction vs. 256 image tokens). Per-token normalization reveals a $38.2\times$ ratio: each instruction token receives $38.2\times$ the attention of each image token in trained instruction-dominant heads, compared to only $1.5\times$ in randomly initialized heads (a $25.5\times$ increase from training). Instruction-dominant heads also show lower attention entropy (0.465) than non-dominant heads (0.569), indicating more concentrated attention distributions.

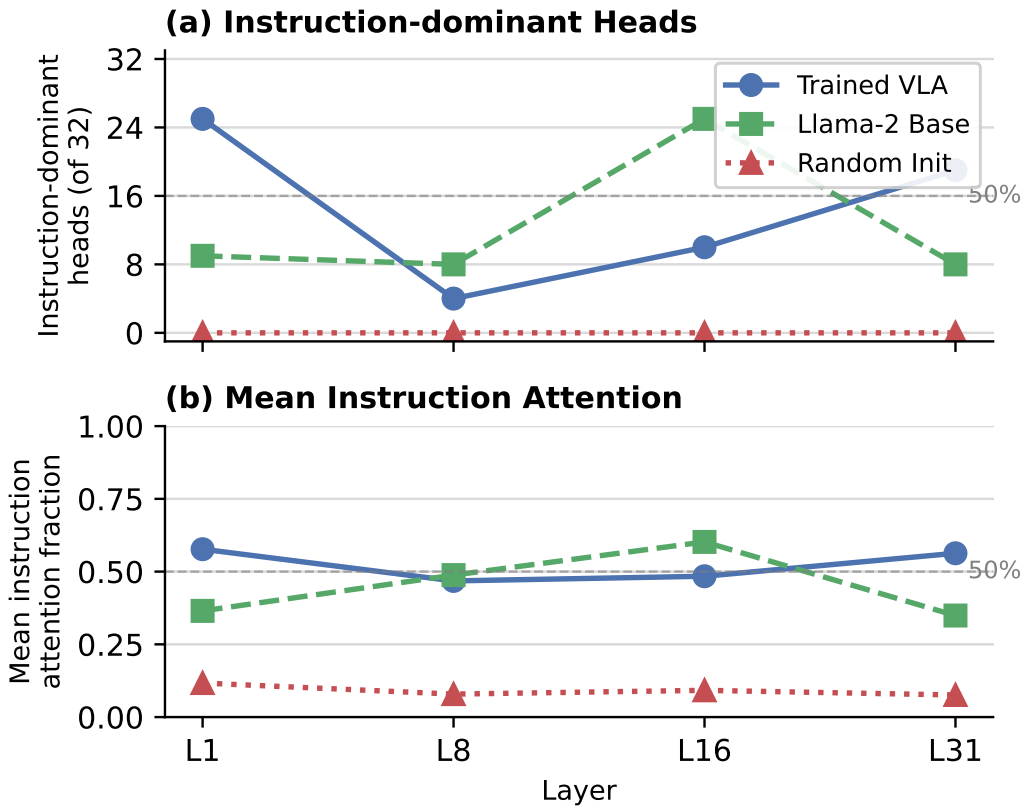


Figure 20: Instruction-dominant heads and mean text attention fraction across layers for three conditions. VLA-trained shows elevated values at L1 and L31; Llama-2 base peaks at L16; randomly initialized remains near zero throughout.

A.24 Per-Dimension Action MAE Under Head Ablation

Table 23 extends Table 8 with full per-dimension control-head data and instruction-to-control MAE ratios.

Table 23: Mean absolute error per action dimension under L1 head ablation. Translation dimensions (dx/dy/dz) show 3–5× larger instruction-to-control ratio than rotation dimensions (1.9–2.6×); instruction-dominant heads have disproportionate influence on positional control. Gripper (2.07×) falls between the two groups.

Dimension	Instr. 24h MAE	Control 8h MAE	Ratio
dx	0.00189	0.00037	5.09×
dy	0.00531	0.00125	4.26×
dz	0.00762	0.00243	3.14×
$roll$	0.02059	0.00785	2.62×
$pitch$	0.03071	0.01387	2.21×
yaw	0.03388	0.01783	1.90×
$gripper$	0.29072	0.14072	2.07×

A.25 Per-Task R^2 Distribution Under L1 Ablation

Table 24 reports per-task action probe R^2 at L1 under baseline and 24-head ablation conditions (MLP probe, 5 demonstrations per task).

Table 24: Per-task action probe R^2 at L1 (MLP probe, 5 demos/task). All 10 tasks show positive ablation effect ($\Delta > 0$): instruction-head ablation uniformly increases action decodability at the entry layer. Mean $\Delta R^2 = +0.107$; the 50-demo setting (§4.2) yields a consistent $+0.059$.

Task	Baseline R^2	Ablated R^2	ΔR^2
open middle drawer	0.197	0.221	+0.024
open top drawer + bowl	0.265	0.340	+0.075
push plate to stove	0.190	0.197	+0.007
put bowl on plate	0.375	0.474	+0.099
put bowl on stove	0.235	0.395	+0.160
put bowl on cabinet	0.256	0.457	+0.201
put cream cheese in bowl	0.186	0.345	+0.159
put wine on rack	0.189	0.424	+0.235
put wine on cabinet	0.279	0.353	+0.074
turn on stove	0.385	0.421	+0.036
Mean $\pm$ std	0.256 ± 0.070	0.363 ± 0.088	+0.107

A.26 Task Complexity and Attention Dependence

We classify tasks as *instruction-discriminative* when multiple tasks share the same manipulated object(s), requiring instruction content to disambiguate the target action (e.g., "put bowl on plate" vs. "put bowl on stove"); remaining tasks involve unique object-action pairings recoverable from visual context alone. Table 25 reports per-task $|\Delta R^2|$ under both groupings.

Table 25: Task-level L1 ablation effects ($|\Delta R^2|$) grouped by instruction dependence. *Primary* (conservative): cream cheese in bowl classified as visual-sufficient ($2.5\times$, $p = 0.033$). *Sensitivity* (shared-object criterion): cream cheese reclassified as instruction-discriminative ($6.5\times$, $p = 0.008$).

Task	$\lvert\Delta R^2\rvert$	Group
put wine on rack	0.235	Instr-disc
put bowl on cabinet	0.201	Instr-disc
put bowl on stove	0.160	Instr-disc
put bowl on plate	0.099	Instr-disc
open top drawer + bowl	0.075	Instr-disc
put wine on cabinet	0.074	Instr-disc
put cream cheese in bowl†	0.159	Vis-suff
turn on stove	0.036	Vis-suff
open middle drawer	0.024	Vis-suff
push plate to stove	0.007	Vis-suff
Primary: Instr-disc mean 0.141; Vis-suff mean 0.057		
Mann-Whitney $U = 21.0$, $p = 0.033$, ratio $= 2.49\times$		
Sensitivity†: cream cheese as Instr-disc: $p = 0.008$, $6.5\times$		

Leave-one-out sensitivity. All leave-one-out p-values remain ≤ 0.083 (marginally significant), with no single task removal reversing the effect direction. Reclassifying the boundary case (cream cheese in bowl) to the instruction-discriminative group strengthens the result ($p = 0.008$).

Bootstrap 95% CI on the complex/simple ratio: $[1.03, 11.12]$ (10,000 resamples). The wide interval reflects low statistical power from $N = 10$ tasks split into groups of 6 and 4; the point estimate ($2.5\times$) persists under leave-one-out analysis (all $p \leq 0.083$, marginally significant; §A.26).

A.27 Specificity Control: Matched-Size Head Ablation

To test whether ablation effects are specific to instruction-dominant heads, we ablate four matched-size subsets of 24 heads at L1: instruction-dominant (highest instruction attention fraction), image-

dominant (highest image attention fraction), random (5 seeds), and bottom-24 (lowest instruction attention fraction; overlaps 16/24 heads with the instruction-dominant group).

Table 26: Matched-size ablation controls: trajectory disruption (d) from ablating 24-head subsets at L1, ranked by instruction attention fraction. Instruction-dominant heads produce the largest effect, but the modest gap (8% over random) indicates distributed, attention-graded redistribution rather than a dedicated subset.

Condition (24 heads)	Cohen's d	95% CI
Instruction-dominant	0.650	(reference)
Random (5 seeds)	0.601 ± 0.021	$[0.568, 0.622]$
Image-dominant	0.580	$[0.547, 0.613]$
Bottom-24	0.565	—

All conditions yield comparable ground-truth MSE (0.487–0.504 vs. baseline 0.497), consistent with direction-selective rather than precision-degrading effects.

A.28 Paired Effect Sizes

Main-text Cohen's d values use the standard (unpaired) formula for token-level comparisons where observations are independent tokens. For within-subject effects where the same tasks or demonstrations are measured under two conditions, we additionally report paired $d_z = \bar{\Delta}/s_\Delta$, which accounts for within-subject correlation and yields larger effect sizes when correlation is high (Table 27).

Table 27: Paired effect sizes (d_z) for within-subject comparisons. Standard (unpaired) d reported where available. Paired d_z exceeds standard d when within-subject correlation is high.

Comparison	Standard d	Paired d_z	N
L1 ablation: 24 dominant vs. baseline	0.65	1.17	10 tasks
Random-24 ablation vs. baseline	—	1.84 ± 0.33	30 subsets
Dose-response: 32 vs. 4 heads (MSE)	—	1.79	10 tasks
RSA: trained vs. reference (best-vs-best)	—	0.55	10 tasks
Cross-arch: OpenVLA vs. TraceVLA peak	—	0.48	15 seeds

A.29 Cross-Architecture Action Decodability: 2×2 Design

Table 29 reports per-layer action prediction R^2 across two architectures (OpenVLA, TraceVLA) $\times$ two training conditions (VLA-trained, reference base LLM), providing a 2×2 factorial view of how per-layer action decodability differs between trained and reference conditions.

Convergent reorganization from divergent starting points. Reference profiles differ qualitatively: OpenVLA (Llama-2) peaks at L0 ($R^2 = 0.629$) then collapses to near-zero at mid-layers (L20: 0.086, std $= 0.311$), while Phi-3-V peaks at L6 (0.578 ± 0.018) with gradual monotonic decline (late-layer std $= 0.068$). After VLA training, both architectures reach nearly identical mid-layer peaks—OpenVLA at L9 ($R^2 = 0.637 \pm 0.008$), TraceVLA at L11 (0.634 ± 0.009)—with $\Delta = 0.003$ and overlapping CIs. Deep-layer variance drops sharply with training: OpenVLA std $0.277 \rightarrow 0.040$; Phi-3-V std $0.068 \rightarrow 0.011$. Peak R^2 values are comparable before and after training (reference peaks reach 91–99% of trained), but peak locations shift deeper (L0→L9; L6→L11) and profile shapes transform from architecture-specific patterns to a shared mid-layer plateau.

Summary. Table 28 condenses the key contrasts. The L0 values are identical between trained and reference within each architecture (shared frozen vision encoder + projector), confirming that training effects arise from LLM weight changes. The largest training gains appear at mid-to-deep layers (L15–L25 average: OpenVLA $+0.313$, Phi-3-V $+0.103$), precisely where reference profiles diverge most.

Table 28: R^2 profile summary across architectures and training conditions.

	OpenVLA (Llama-2-7B)		TraceVLA (Phi-3-mini)	
	Reference	Trained	Reference	Trained
Peak Layer	L0	L9	L6	L11
Peak R^2	0.629	0.637	0.578	0.634
Profile Shape	Collapse	Flat plateau	Ramp $\rightarrow$ decline	Ramp $\rightarrow$ plateau
Late-layer std	0.277	0.040	0.068	0.011
Δ_{train} (peak)	$+0.008$ (L0$\rightarrow$L9)		$+0.056$ (L6$\rightarrow$L11)	
Δ_{train} (L15–L25 avg)	$+0.313$		$+0.103$	

Equivalence testing. To formally test whether the cross-architecture peak R^2 values are statistically equivalent (rather than merely failing to differ), we apply the Two One-Sided Tests (TOST) procedure [53] with equivalence bound $\varepsilon = 0.05$. The observed difference is $\Delta = 0.003$ (OpenVLA L9: 0.637 ± 0.008; TraceVLA L11: 0.634 ± 0.009). Both one-sided tests reject at $p < 0.0001$; the 90% confidence interval for the difference is $[-0.002, 0.009]$, falling entirely within $[-0.05, 0.05]$. A conventional two-sample t-test yields $p = 0.318$ (not significant), illustrating the distinction between absence of evidence for a difference and evidence of equivalence: the TOST procedure provides the latter.

A.30 Global Multiple Comparison Correction

The paper reports 63 statistical tests spanning permutation tests, t-tests, bootstrap comparisons, TOST equivalence, and Spearman correlations. We apply a Bonferroni correction ($\alpha_{\text{adj}} = 0.05/63 = 7.94 \times 10^{-4}$) and the less conservative Holm–Bonferroni step-down procedure; both yield identical results. Of the 45 originally significant tests, 40 survive correction (88.9%).

All eight core tests supporting the paper's main claims remain significant: RSA permutation at L12 ($p < 0.0001$), trained vs. reference image geometry at L12 ($p = 3.9 \times 10^{-5}$), counterfactual wrong and empty instruction ($p < 10^{-100}$), TOST cross-architecture equivalence ($p < 0.0001$), dose-response top-24 ablation ($p < 10^{-100}$), trained vs. reference probe R^2 ($p < 0.0001$), and bottom-24 specificity ($p < 10^{-90}$).

Five auxiliary tests lose significance under correction: (i) RSA trained vs. reference preaction at L12 ($p = 0.005 \rightarrow p_{\text{adj}} = 0.284$), (ii) DINOv2 vs. SigLIP RSA ($p = 0.032 \rightarrow p_{\text{adj}} = 1.0$), (iii) fused backbone vs. SigLIP RSA ($p = 0.011 \rightarrow p_{\text{adj}} = 0.673$), (iv) CV exact paired permutation ($p = 0.016 \rightarrow p_{\text{adj}} = 0.984$), and (v) R^2 projector vs. VLA Spearman correlation ($p = 0.003 \rightarrow p_{\text{adj}} = 0.183$). These concern auxiliary comparisons (encoder ranking, coefficient-of-variation ordering) rather than the core redistribution, probing, or counterfactual findings; no main-text conclusion requires revision.

A.31 Mechanistic Interpretation

Matched-size ablation (d: instruction 0.65, random 0.60, image 0.58; Table 26) and dose-response (steep through 12 heads, plateau beyond 20) indicate distributed attention redistribution. Counterfactual perturbation produces large effects across all conditions ($d = 1.0$–1.2 on Goal; §4.2), with shuffled tokens exceeding paraphrased ($d = 1.096$ vs. 0.996) despite identical token sets, indicating positional sensitivity. That ablation raises R^2 yet alters trajectories ($d = 0.65$) without increasing ground-truth error ($\Delta\text{MSE} \approx 0$; §A.12) reflects an information-computation dissociation [54]: instruction-dominant heads control direction rather than precision. Random ablation yields larger R^2 gains ($+0.071$ vs. $+0.030$; $z = -3.11$, 0th percentile), consistent with distinct R^2 signatures.

Table 29: Per-layer action prediction R^2: training effect across architectures (mean $\pm$ std over 15 seeds). Peak values per condition in **bold**.

Layer	OpenVLA (Llama-2-7B)		TraceVLA (Phi-3-mini)	
	Reference	Trained	Reference	Trained
L0	$\mathbf{0.629 \pm 0.016}$	0.629 ± 0.016	-0.330 ± 0.003	-0.330 ± 0.002
L1	0.539 ± 0.023	0.625 ± 0.017	0.429 ± 0.034	0.447 ± 0.013
L2	0.517 ± 0.047	0.622 ± 0.010	0.458 ± 0.023	0.522 ± 0.014
L3	0.448 ± 0.216	0.624 ± 0.014	0.498 ± 0.036	0.579 ± 0.011
L4	0.496 ± 0.057	0.628 ± 0.014	0.572 ± 0.021	0.592 ± 0.009
L5	0.475 ± 0.101	0.627 ± 0.019	0.573 ± 0.020	0.602 ± 0.010
L6	0.422 ± 0.119	0.631 ± 0.012	$\mathbf{0.578 \pm 0.018}$	0.609 ± 0.012
L7	0.459 ± 0.093	0.632 ± 0.012	0.569 ± 0.023	0.616 ± 0.014
L8	0.347 ± 0.283	0.635 ± 0.013	0.560 ± 0.025	0.615 ± 0.010
L9	0.445 ± 0.127	$\mathbf{0.637 \pm 0.008}$	0.564 ± 0.021	0.617 ± 0.013
L10	0.353 ± 0.224	0.624 ± 0.014	0.551 ± 0.027	0.625 ± 0.007
L11	0.356 ± 0.122	0.623 ± 0.016	0.561 ± 0.025	$\mathbf{0.634 \pm 0.009}$
L12	0.409 ± 0.203	0.626 ± 0.011	0.557 ± 0.026	0.632 ± 0.005
L13	0.341 ± 0.227	0.624 ± 0.013	0.547 ± 0.028	0.633 ± 0.008
L14	0.374 ± 0.211	0.623 ± 0.015	0.550 ± 0.023	0.631 ± 0.009
L15	0.297 ± 0.225	0.619 ± 0.012	0.550 ± 0.022	0.633 ± 0.010
L16	0.323 ± 0.226	0.619 ± 0.011	0.534 ± 0.018	0.631 ± 0.010
L17	0.395 ± 0.109	0.609 ± 0.023	0.519 ± 0.024	0.633 ± 0.007
L18	0.214 ± 0.307	0.613 ± 0.017	0.502 ± 0.047	0.629 ± 0.012
L19	0.241 ± 0.258	0.613 ± 0.013	0.499 ± 0.058	0.627 ± 0.008
L20	0.086 ± 0.311	0.611 ± 0.014	0.482 ± 0.048	0.628 ± 0.011
L21	0.330 ± 0.216	0.589 ± 0.023	0.476 ± 0.036	0.624 ± 0.007
L22	0.207 ± 0.283	0.586 ± 0.013	0.408 ± 0.140	0.616 ± 0.008
L23	0.264 ± 0.227	0.578 ± 0.028	0.455 ± 0.074	0.613 ± 0.007
L24	0.173 ± 0.275	0.577 ± 0.013	0.446 ± 0.062	0.610 ± 0.008
L25	0.130 ± 0.305	0.553 ± 0.053	0.479 ± 0.039	0.609 ± 0.010
L26	0.251 ± 0.254	0.535 ± 0.046	0.411 ± 0.098	0.611 ± 0.008
L27	0.099 ± 0.278	0.552 ± 0.027	0.366 ± 0.197	0.604 ± 0.011
L28	0.099 ± 0.307	0.544 ± 0.034	0.358 ± 0.218	0.596 ± 0.015
L29	0.088 ± 0.275	0.530 ± 0.034	0.356 ± 0.271	0.585 ± 0.010
L30	0.220 ± 0.234	0.511 ± 0.029	0.424 ± 0.104	0.574 ± 0.008
L31	0.069 ± 0.284	0.488 ± 0.054	0.332 ± 0.262	0.557 ± 0.016
L32	0.584 ± 0.011	0.535 ± 0.015	0.410 ± 0.185	0.550 ± 0.007

A.32 Functional Stages and Predictions

Per-layer profiles suggest three stages: redistribution (L0–L1; probe 99.6%, 24/32 instruction-dominant heads), computation (L8–L13; R^2 peak 0.655–0.678), and decay (L14–L31, $\sim 56\%$ of parameters; declining R^2 and RSA suggest pruning potential). Task-specific fine-tuning eliminates L31 re-engagement ($18 \rightarrow 0$ heads; §4.2), consistent with exit-layer attention serving multi-task flexibility (N=2). TraceVLA [7] (Phi-3-V) shows consistent entry-exit attention concentration from an opposite starting point (§4.2). The 2×2 trained-vs-reference comparison reveals a consistent reorganization pattern: reference OpenVLA collapses from L0 ($R^2 = 0.629$) to near-zero at mid-layers (L20: 0.086, std $= 0.277$), while reference Phi-3-V peaks at L6 (0.578) with gradual decline (std $= 0.068$)—qualitatively opposite reference landscapes. Yet training drives both to nearly identical

mid-layer peaks (0.637 vs. 0.634, $\Delta = 0.003$) with stabilized deep-layer variance (OpenVLA std $0.277 \rightarrow 0.040$; Phi-3-V $0.068 \rightarrow 0.011$). This consistent outcome from divergent starting points suggests that mid-layer action specialization is a consistent convergence pattern of VLA training, not an architectural artifact. Exit-layer R^2 differs (0.488 vs. 0.557), possibly reflecting architectural differences in late-layer processing. Remaining predictions: cross-attention VLAs (π_0 [14], Octo [51]) should lack L1 instruction-dominant heads; larger VLAs should yield flatter per-head contributions. For deployments, L1 instruction probes can flag instruction-discriminative tasks and L1 attention entropy can monitor instruction under-attendance at runtime.

A.33 Extended Discussion

Observational three-tier gradient. The cross-architecture matrix suggests a three-tier gradient of representational adaptation. Tier 1 (VLM bases, no action training): inherited L1 attention (TracVLA 32/32, NORA 13–14/16), negligible exit collapse ($\Delta R^2 \leq 0.048$), and minimal counterfactual sensitivity (cosine distance ≤ 0.185). Tier 2 (general-purpose VLA, OXE-pretrained): strong L1 routing (24–26/32 heads), moderate counterfactual sensitivity ($d \approx 1.0$), and no systematic exit collapse across suites. Tier 3 (task-specific VLAs, fine-tuned on LIBERO): exit collapse at every VLM-to-VLA cell and medium-to-large counterfactual sensitivity ($d = 0.45$–1.52). This gradient is suggestive but limited: Tier 2 contains a single architecture (OpenVLA), and the Tier 2/Tier 3 boundary confounds training stage with architecture. Testing additional general-purpose VLAs (e.g., Octo [51], π_0 [14]) at Tier 2 is necessary before the three-tier structure can be treated as a robust empirical pattern.

Exit collapse mechanism. One possible explanation for the VLM-to-VLA / VLA-to-VLA asymmetry is that VLM-to-VLA references lack action-relevant exit-layer structure, forcing mid-layer construction without exit preservation (analogous to fine-tuning-induced feature distortion [19]), while VLA-to-VLA references already maintain this structure. Validating this explanation causally requires controlled training-stage experiments, but the consistency across architectures supports its directional value.

Behavioral relevance. Correlating trained ΔR^2 with published LIBERO success rates across all nine available architecture-suite cells yields a positive but non-significant trend (Spearman $\rho = 0.583$, $p = 0.10$, $n = 9$), marginally above the conventional significance threshold. Within-suite comparisons, which control for task difficulty, are uniformly positive: on Goal ($n = 3$ architectures), ΔR^2 and success rate are perfectly rank-correlated ($\rho = 1.0$), and every remaining suite ($n = 2$ each) shows the same direction. This directional evidence is consistent with stronger mid-network action encoding (higher peak R^2) followed by more aggressive exit-layer compression being associated with better task performance. Several caveats temper this interpretation: architecture capacity differences (model size, backbone quality) are not controlled, suite difficulty co-varies with both ΔR^2 and success rate, multiple suites from the same architecture are not independent observations, and the sample size limits statistical power. This correlational evidence does not establish a causal link between exit collapse and task success. Closed-loop behavioral validation, including intervention experiments that directly manipulate exit-layer representations, remains a priority for future work.

Metric constraints. Counterfactual sensitivity is measured via Cohen's d on action MSE for models that produce actions and via cosine distance on hidden representations for reference VLMs (§4.4). These metrics differ in scale and distributional properties; ratios between them are not meaningful. The OpenVLA Long-Horizon inversion (trained $d = 0.56$, reference $d = 1.05$) illustrates how same-metric comparisons can reverse the expected direction, cautioning against overinterpretation of any single cell. A standardized metric framework for cross-condition counterfactual comparison remains an open problem.

Extended limitations. Each architecture is analyzed at a single trained checkpoint, precluding characterization of how exit collapse or counterfactual sensitivity develop during training. All analyses are open-loop, consistent with prior representational studies [5, 6]; closed-loop validation would provide complementary evidence about the behavioral relevance of the identified routing structures. The three-tier gradient has $n = 1$ at Tier 2, limiting its inferential power; populating this tier with additional general-purpose VLAs is a priority for future work.

A.34 NLP Finetuning Controls

Table 30 compares the three NLP finetuning controls used to test whether exit collapse is specific to action-oriented training. All three chat-finetuned models produce flat action-probe profiles with $\Delta R^2 = 0.000$, confirming that NLP instruction tuning does not induce the peak-to-exit drop observed in VLM-to-VLA transitions in any of the three backbone families.

Table 30: **NLP finetuning control comparison.** All three models are instruction-tuned for conversational use without action heads or robot demonstration data. None exhibits exit collapse.

Model	VLA backbone	ΔR^2	Mean R^2	Layers
Llama-2-7b-chat	OpenVLA (Llama-2-7B)	0.000	0.155 ± 0.001	32
Phi-3-mini-4k-instruct	TracVLA (Phi-3-mini)	0.000	0.155 ± 0.000	32
Qwen2.5-3B-Instruct*	NORA (Qwen2.5-VL-3B)	0.000	0.155 ± 0.000	36

*Qwen2.5-3B-Instruct shares the Qwen2.5 language backbone with NORA's Qwen2.5-VL-3B but lacks the vision encoder and cross-attention modules.

A.35 Claim-Evidence Summary

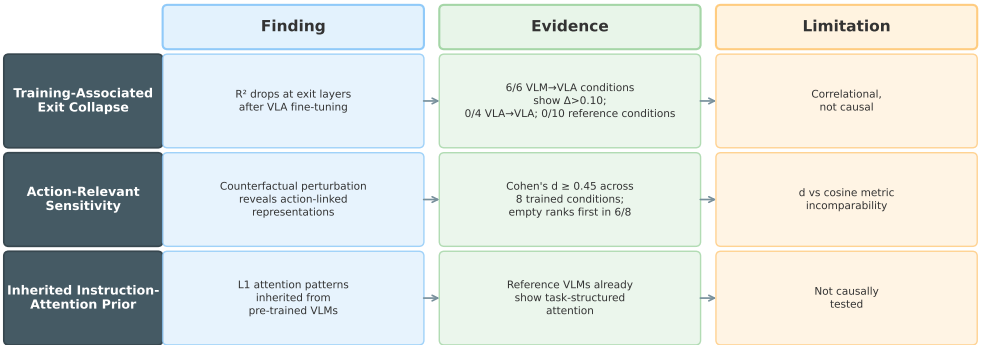


Figure 21: **Claim–evidence–scope summary.** Each main finding is paired with its supporting analysis and its main limitation. The figure distinguishes directly observed representational patterns from stronger causal interpretations that require interventional or training-dynamics evidence.